

Peripheral Nerves and Skeletal Muscles

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Neuromuscular diseases are caused by genetic, metabolic, degenerative, or inflammatory conditions that may be inherited or acquired. They typically present with weakness, muscle pain, or sensory deficits and can be grouped according to anatomy, pattern of injury, disease course, and pathogenesis. Physicians consider all these characteristics when evaluating a patient with neuromuscular symptoms. This chapter uses an anatomic approach, grouping neuromuscular disorders into those that preferentially affect the peripheral nerves, the neuromuscular junction, or the skeletal muscles. A discussion of neoplasms that arise from peripheral nerves ends the chapter. Conditions that produce similar clinical symptoms but are caused by disorders of the central nervous system (CNS) are discussed in Chapter 28.

DISEASES OF PERIPHERAL NERVES

The two components of peripheral nerves involved in impulse transmission are axons that arise from neurons and myelin sheaths made by Schwann cells, which form the protective covering around nerves. Injury to either of these components may result in a peripheral neuropathy.

Before discussing the pathology of these disorders, a brief review of peripheral nerve structure and function is in order. *Somatic motor function* is carried out by the motor unit, which consists of (1) a lower motor neuron located in the anterior horn of the spinal cord or in the brainstem; (2) an axon that travels in a nerve to a target; (3) the neuromuscular junction; and (4) multiple innervated myofibers (muscle fibers). *Somatic sensory function* depends on (1) the distal nerve endings, which may contain specialized structures that serve to register specific sensory modalities; (2) a distal axon segment that travels as part of a peripheral nerve to the dorsal root ganglion; and (3) a proximal axon segment that synapses on neurons in the spinal cord or brainstem. *Autonomic nerve fibers*, which transmit all visceral motor and sensory functions, outnumber somatic fibers in the peripheral nervous system, but signs and symptoms related to their involvement are generally not prominent features of peripheral neuropathies, with a few important exceptions (e.g., in some cases of diabetic neuropathy, discussed later).

Specific sensations (pain, temperature, touch) and motor signals are each conveyed by axons that can be distinguished, in part, based

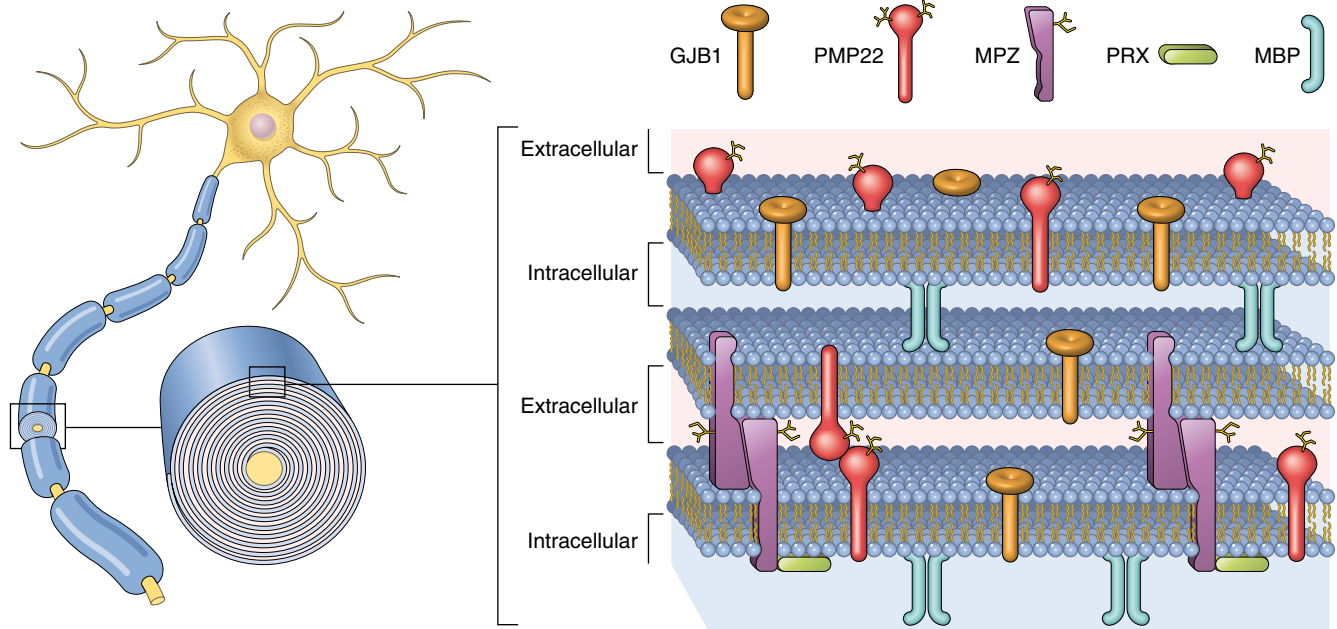


Fig. 27.1 Relationship between lipid bilayers and associated proteins in myelin within internodes. Myelin basic protein (MBP) is an intracellular protein that has a role in myelin compaction. Mutant forms of myelin protein zero (MPZ), peripheral myelin protein 22 (PMP22), gap junction protein beta 1 (GJB1), and periaxin (PRX) cause some forms of Charcot-Marie-Tooth disease, a hereditary demyelinating and/or axonal neuropathy, and other rare genetic neuropathies.

on their diameter. Axonal diameters are in turn correlated with the thickness of their myelin sheaths and conduction speeds:

- *Thin, unmyelinated fibers* mediate autonomic functions as well as certain pain and temperature sensations and have the slowest conduction speeds due to their lack of myelin and small axonal diameter.
- *Large diameter axons* with thick myelin sheaths transmit light touch, proprioception, and motor signals and have fast conduction speeds.

In the case of myelinated axons, individual Schwann cells make and maintain exactly one myelin sheath that wraps around a single axon to create a myelinated segment called an *internode*. Internodes are separated by unmyelinated gaps referred to as *nodes of Ranvier*, which are uniformly spaced along the length of the axon and allow for saltatory conduction. A number of specialized proteins are essential for normal assembly and function of myelin within internodes (Fig. 27.1). Unmyelinated axons are also intimately associated with Schwann cells but in a different arrangement in which one cell surrounds segments of multiple axons.

Most peripheral nerves carry out both motor and sensory functions and thus contain axons of varying diameter and myelin thickness. The axons are bundled together by three major connective tissue components: the epineurium, fibrovascular tissue that encloses the entire nerve; the perineurium, a multilayered concentric specialized connective tissue sheath that groups subsets of axons into fascicles; and the endoneurium, which surrounds individual nerve fibers.

General Types of Peripheral Nerve Injury

Axonal Neuropathies

Axons are the primary target of the damage in this large group of peripheral neuropathies (Fig. 27.2). The morphologic hallmarks of axonal neuropathies can be produced experimentally by cutting a peripheral nerve, which results in a prototypical pattern of injury described as *Wallerian degeneration*. Portions of axons that are distal to the point of transection are disconnected from the cell bodies (perikarya) and

degenerate. Within a day of injury, the distal axons begin to fragment, and the associated myelin sheaths unravel (Fig. 27.3) and disintegrate into spherical structures (*myelin ovoids*). Macrophages are recruited and participate in the removal of axonal and myelin debris. Regeneration starts at the site of transection with the formation of a growth cone, followed by the outgrowth of new branches from the stump of the proximal axon. Schwann cells and their associated basement membranes guide the sprouting axons, which grow at about 1–2 mm per day toward their distal target. Continuous pruning of the sprouting axons removes misguided branches. The Schwann cells create new myelin sheaths around the regenerating axons, but these myelin internodes tend to be thinner and shorter than in the original ones. The regeneration is successful only if the two transected ends remain closely approximated. A failure of the outgrowing axons to find their distal target can produce a “pseudotumor” termed *traumatic neuroma*—a nonneoplastic haphazard proliferation of axonal processes and associated Schwann cells that results in a painful nodule (Fig. 27.4).

The changes observed following experimental nerve transections only partially resemble those seen in various axonal neuropathies. One key difference is that in these neuropathies (unlike nerve transection), damage occurs over an extended period of time. As a result, *degenerating and regenerating axons coexist in a single lesion*. With time, damage tends to outpace repair, resulting in progressive loss of axons. In cases of toxic and metabolic insults, axons often degenerate in a length-dependent fashion with the longest axons being most susceptible, resulting in a “dying-back” pattern of progression. The electrophysiologic hallmark of axonal neuropathies is a reduction in signal amplitude owing to the dropout of axons from affected peripheral nerves, with relative preservation of conduction velocity.

Demyelinating Neuropathies

In these disorders, Schwann cells and their myelin sheaths are the primary target of damage (Fig. 27.2), whereas axons are relatively preserved. This definition is similar to that of demyelinating diseases

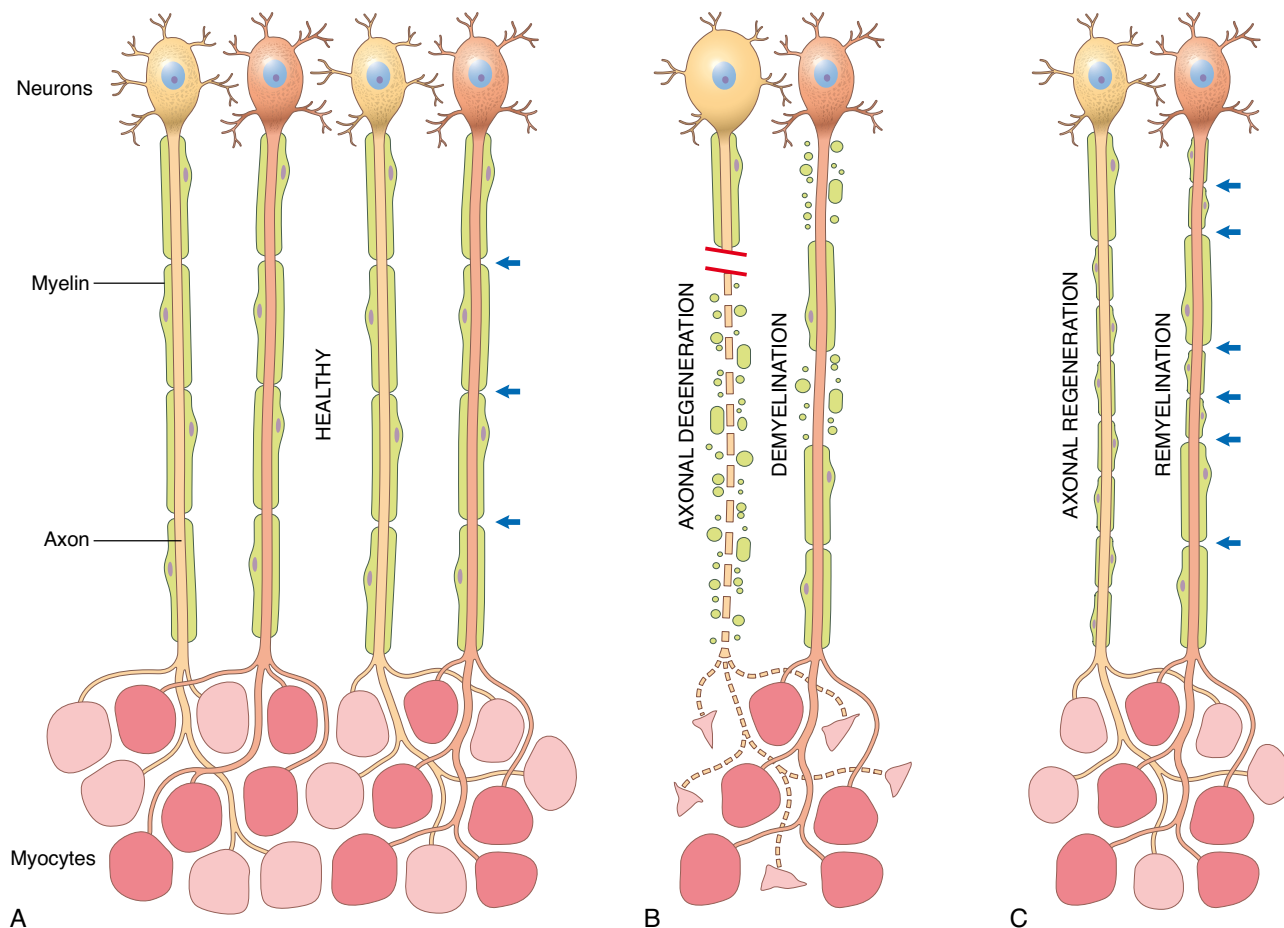


Fig. 27.2 Patterns of peripheral nerve damage. (A) In normal motor units, type I and type II myofibers are arranged in a “checkerboard” distribution. The internodes, separated by nodes of Ranvier (arrows) along the motor axons are uniform in thickness and length. (B) Acute axonal injury (*left axon*) results in degeneration of the distal axon and its associated myelin sheath, with angular atrophy of denervated myofibers. In contrast, acute demyelinating disease (*right axon*) produces random segmental demyelination of individual internodes, while sparing the axons. (C) Regeneration of axons after injury (*left axon*) allows reinnervation of myofibers. The regenerated axon is myelinated by proliferating Schwann cells, but the new internodes are shorter and the myelin sheaths are thinner than the original ones. Remyelination of demyelinating disease (*right axon*) allows remyelination to take place, but the new internodes are shorter and have thinner myelin sheaths than flanking normal undamaged internodes (Nodes of Ranvier are marked by arrows; compare with panel A). See Fig. 27.7 for comparison with reinnervation.

that affect the CNS (Chapter 28). The pathogenesis of demyelinating peripheral neuropathies is most often immunologic or hereditary. Individual myelin sheaths are damaged in a seemingly random pattern, resulting in discontinuous demyelination of internodes. In response to this damage, Schwann cells or their precursors proliferate and initiate repair through the formation of new myelin sheaths, but these tend to be shorter and thinner than the original ones. The electrophysiologic hallmark of demyelinating neuropathies is slowed nerve conduction velocity, reflecting the loss of myelin and its effect on saltatory conduction; however, amplitude is preserved.

Neuronopathies

Neuronopathies result from destruction of neurons, leading to secondary degeneration of axonal processes and subsequently of myelin. Infections like herpes zoster and toxins like platinum compounds are examples of insults that may lead to neuronopathies. Inherited and

degenerative forms of neuronopathy can also occur. Because the damage is at the level of the neuronal cell body, peripheral nerve dysfunction caused by neuronopathies is equally likely to affect proximal and distal parts of the body (unlike peripheral axonopathies, which preferentially affect the distal extremities).

Anatomic Patterns of Peripheral Neuropathies

Peripheral neuropathies can be separated into several groups according to the anatomic distribution of involvement and the associated neurologic deficits. This approach can be helpful clinically, since each pattern has a different set of potential underlying causes. The anatomic patterns of injury are:

- **Mononeuropathies** affect a single nerve and result in deficits in a restricted distribution dictated by nerve anatomy. Trauma, entrapment/compression, and infections are common causes of mononeuropathy.

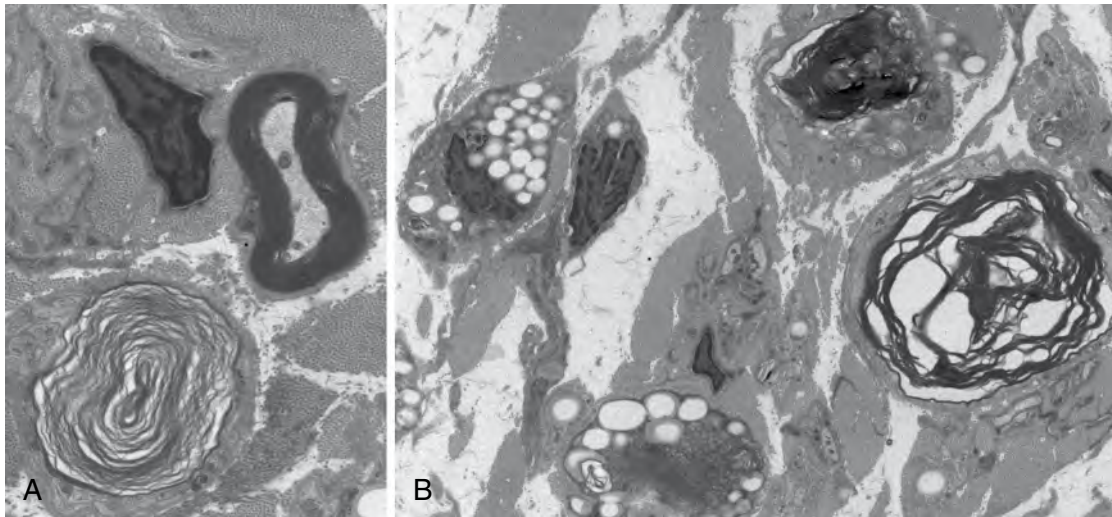


Fig. 27.3 Electron micrographs illustrating features of axonal degeneration. (A) Active axonal degeneration with loosened myelin layers is seen in the axon in the lower left corner, to be contrasted with a normal myelin sheath with tightly packed myelin surrounding an intact axon in the upper right corner. (B) In addition to an unraveling myelin sheath, several cells contain lipid droplets (seen as vacuoles) derived from degenerating myelin.

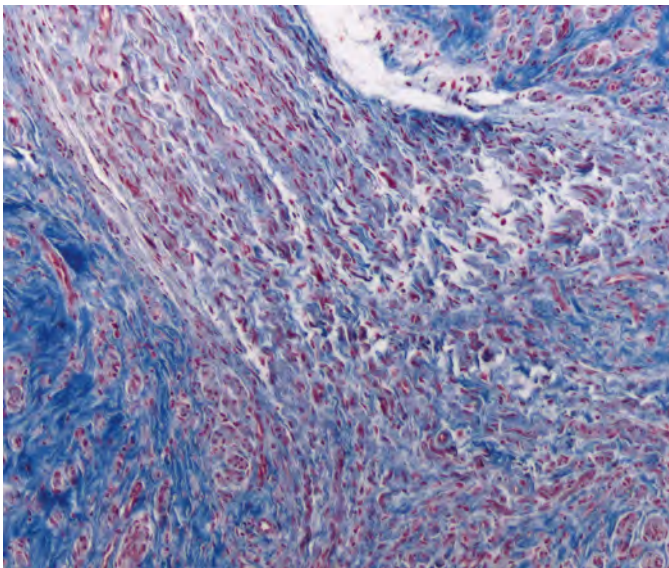


Fig. 27.4 Traumatic neuroma. Masson trichrome-stained section showing the transition from normal nerve containing a parallel arrangement of axons (*upper left corner*) to a haphazard swirl of red-stained axons associated with admixture of Schwann cells and blue-staining connective tissue.

- **Polyneuropathies** are characterized by involvement of multiple nerves, usually in a symmetric fashion. In most cases axons are affected in a length-dependent fashion leading to deficits that start in the feet and ascend with disease progression. The hands often start to show involvement by the time deficits extend to the level of the knee, resulting in a characteristic “stocking and glove” distribution of sensory deficits. Metabolic diseases, such as diabetes mellitus, are a common cause of polyneuropathy.
- **Mononeuritis multiplex** describes a disease process that damages individual nerves in a haphazard fashion. An affected patient might have a right wrist drop from involvement of the right radial nerve

and a left foot drop from peroneal nerve damage. Vasculitis is a common cause of this pattern of injury.

- **Polyradiculoneuropathies** affect nerve roots as well as peripheral nerves, leading to diffuse symmetric symptoms in proximal and distal parts of the body. Autoimmune causes are common such as chronic inflammatory demyelinating polyradiculoneuropathy (CIDP).

Specific Peripheral Neuropathies

Patients with peripheral neuropathy often complain of numbness, painful “pins and needles” sensations, and weakness, most often in the distal portions of the extremities. Many different disease processes can damage peripheral nerves, including inflammatory or immune-mediated diseases, infections, metabolic changes, toxic injury, trauma, paraneoplastic disease, and inherited or *de novo* gene defects. The peripheral neuropathies discussed below are organized based on pathogenesis; different neuropathies also show different patterns of injury ([Table 27.1](#)).

Inflammatory Neuropathies

Guillain-Barré Syndrome. Guillain-Barré syndrome (GBS), the most common form of acute inflammatory demyelinating polyradiculoneuropathy (AIDP), is an immune-mediated demyelinating peripheral neuropathy that may lead to life-threatening respiratory paralysis. The overall annual incidence is approximately one case per 100,000 persons. The disease is characterized by weakness beginning in the distal limbs that rapidly advances to affect proximal muscle function (*ascending paralysis*). Histologic features are inflammation and demyelination of spinal nerve roots and peripheral nerves. Because spinal nerve roots are affected in addition to peripheral nerves, the pattern of injury in GBS is best classified as a radiculoneuropathy.

Pathogenesis

The peripheral nerve injury in GBS is believed to be caused by an autoimmune response likely triggered by exposure to a microbe or microbial antigen. Approximately two-thirds of cases are preceded by an acute upper respiratory illness from which the affected individual

TABLE 27.1 Diseases of Peripheral Nerve

Disorder	Principal pathologic manifestations
Inflammatory/Immune Mediated	
Guillain-Barré syndrome (acute inflammatory demyelinating polyradiculoneuropathy)	Segmental demyelination/remyelination
Chronic inflammatory demyelinating polyradiculoneuropathy	Segmental demyelination/remyelination
Vasculitis	
Autoimmune diseases (such as rheumatoid arthritis, Sjögren syndrome, systemic lupus erythematosus)	Mixed axonal degeneration and demyelination
Infections	
Diphtheria	Segmental demyelination/remyelination
Lyme disease, HIV/AIDS, VZV	Axonal degeneration/regeneration
Leprosy	Mixed axonal degeneration and demyelination
Malignancy Associated	
POEMS syndrome	Segmental demyelination/remyelination
Radiation therapy	Segmental demyelination/remyelination
Direct infiltration or compression by neoplasm	Axonal degeneration/regeneration
Paraneoplastic	Axonal degeneration/regeneration
Metabolic Diseases	
Diabetes, uremia, hypothyroidism, nutritional deficiency (including B6, B1, folate, copper, zinc)	Axonal degeneration/regeneration
Vitamin B12 deficiency	Mixed axonal degeneration and demyelination
Physical Forces	
Laceration, avulsion, compression	Axonal degeneration/regeneration
Genetic Diseases	
Hereditary sensory neuropathies with or without autonomic neuropathy	Axonal degeneration/regeneration
Familial amyloidosis	Axonal degeneration/regeneration
Charcot-Marie-Tooth disease	Variable axonal degeneration and/or demyelination
Toxic Exposures	
Alcohol, heavy metals, solvents, illicit drugs, and chemotherapeutic agents	Variable axonal degeneration and/or demyelination

Peripheral neuropathies are separated by etiology and grouped based on the primary pattern of damage being demyelinating, axonal, and variable or mixed demyelinating and axonal.

POEMS, Polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy, and skin changes; VZV, varicella-zoster virus.

has recovered by the time the neuropathy becomes symptomatic. Infections with *Campylobacter jejuni*, cytomegalovirus, Epstein-Barr virus, *Mycoplasma pneumoniae*, Zika virus, and SARS-CoV-2 have significant epidemiologic associations with GBS. Very rarely, GBS has been reported in patients after recent vaccination for influenza, varicella-zoster, and SARS-CoV-2 (non-mRNA-based vaccine). No infectious agent has been demonstrated in affected nerves, and an immunologic reaction is favored as the underlying cause. There is a rare axonal type of GBS (acute motor axonal neuropathy; AMAN) showing degeneration/regeneration instead of demyelination/remyelination that has a stronger association with antecedent *Campylobacter jejuni* infection. An inflammatory disease of peripheral nerves similar to the typical demyelinating form of GBS can be reproduced in experimental animals by immunization with a peripheral nerve myelin protein. A T cell-mediated immune response ensues, accompanied by segmental demyelination induced by the actions of activated macrophages. Transfer of these T cells to a naive animal results in comparable lesions. Moreover, lymphocytes from individuals with GBS have been shown to produce demyelination co-cultured with myelinated nerve fibers. Circulating antibodies that cross-react with components of peripheral nerves may also play a role.

MORPHOLOGY

In Guillain-Barré syndrome, the dominant finding in sections stained with hematoxylin and eosin is **inflammation of peripheral nerves**, manifested as perivascular and endoneurial infiltration by macrophages, variable lymphocytes, and few plasma cells. **Segmental demyelination and remyelination** affecting peripheral nerves is the most prominent lesion, but damage to axons is also seen, presumably secondary particularly when the disease is severe. Electron microscopy has identified an early injury involving **myelin sheaths**, in which cytoplasmic processes of macrophages penetrate the basement membrane of Schwann cells, particularly in the vicinity of the nodes of Ranvier, and extend between the myelin lamellae, stripping the myelin sheath from the axon. Ultimately, the remnants of the myelin sheath are engulfed by macrophages. Inflammation and demyelination/remyelination can be widespread in the peripheral nervous system but are typically most prominent proximally, close to the nerve roots.

Clinical Features

The clinical picture of GBS is dominated by ascending paralysis and areflexia with a nadir of weakness reached within 4 weeks of disease onset. Deep tendon reflexes disappear early in the process. Sensory

involvement, including loss of pain sensation, is often present but is usually not a prominent feature. Nerve conduction velocities are slowed because of multifocal destruction of myelin segments in many axons within a nerve. Cerebrospinal fluid (CSF) protein levels are elevated due to inflammation and altered permeability of the microcirculation within the spinal roots as they traverse the subarachnoid space. Inflammatory cells, on the other hand, remain confined to the roots; therefore, there is little or no CSF pleocytosis. Many patients spend weeks in the intensive care unit before recovering normal function. With improved supportive respiratory care, cardiovascular monitoring, and prophylaxis against deep venous thrombosis, the mortality rate has fallen. Plasmapheresis and intravenous immunoglobulin infusions hasten recovery, apparently because these remove pathogenic antibodies and suppress immune function, respectively. However, 2% to 5% of affected patients die of respiratory paralysis, autonomic instability, cardiac arrest, or related complications, and up to 20% of hospitalized survivors have long-term disability.

Chronic Inflammatory Demyelinating Polyradiculoneuropathy (CIDP).

This is the most common chronic, acquired, inflammatory peripheral neuropathy, characterized by symmetric mixed sensorimotor polyneuropathy that persists for 8 weeks or more. By definition, signs and symptoms must be present for at least 2 months, but often the disease evolves over years, usually with relapses and remissions. Most commonly there is a symmetric, mixed sensorimotor polyneuropathy characterized by weakness, difficulty walking, numbness, and pain, but some patients may present with isolated sensory or motor impairment. Clinical remissions can often be achieved with intravenous immunoglobulin infusions, or with other immunosuppressive therapies, such as plasmapheresis, glucocorticoids, and cytotoxic agents directed against T cells or B cells. The longer time course and the positive response to steroids distinguish CIDP from Guillain-Barré syndrome.

Pathogenesis

Both T cell–mediated and antibody-mediated mechanisms directed against molecules expressed at the Schwann cell–axon junction and in compact areas of myelin are implicated in the inflammatory process of CIDP. Complement-fixing immunoglobulin G (IgG) and IgM can be found on the myelin sheaths, and the deposition of these antibodies leads to recruitment of macrophages that strip myelin from axons.

MORPHOLOGY

Sural nerve biopsies show evidence of recurrent demyelination and proliferation of Schwann cells attempting to regenerate myelin that leads to remyelination. When excessive, this proliferation leads to the formation of so-called **onion bulbs**—structures in which multiple layers of Schwann cells wrap around an axon like the layers of an onion (Fig. 27.5). Onion bulb formation is not specific to a diagnosis of CIDP and can be found in any disease with repetitive demyelination/remyelination. With the exception of onion bulb formation, the pathologic findings of CIDP are identical to those seen in GBS, including prominent macrophage infiltration.

Newer research has focused on the identification of peripheral neuropathies with autoantibodies reacting with nodal and paranodal proteins associated with noncompact myelin. These were originally classified as subtypes of CIDP due to some clinical and pathologic overlap but are best categorized separately as autoimmune nodopathies.

Neuropathy Associated With Systemic Autoimmune Diseases. Systemic autoimmune diseases like rheumatoid arthritis, Sjögren syndrome, or systemic lupus erythematosus may be associated with peripheral neuropathies that often take the form of distal sensory or sensorimotor

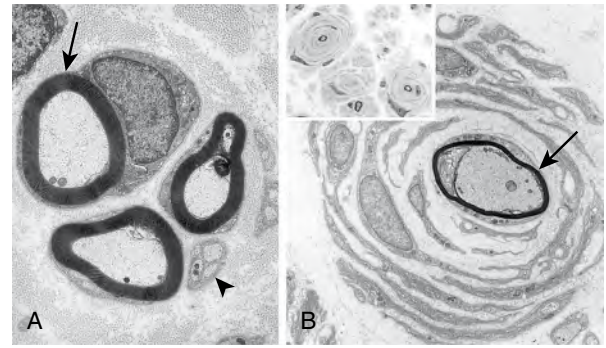


Fig. 27.5 Chronic demyelinating neuropathy. (A) Normal myelinated axons (one axon is associated with the nucleus of a Schwann cell, arrow) and several unmyelinated axons (arrowhead). (B) An “onion bulb” is composed of a thinly myelinated axon (arrow) surrounded by multiple concentrically arranged Schwann cells indicative of repetitive demyelination/remyelination. *Inset*, Light microscopic appearance of a chronic demyelinating neuropathy characterized by “onion bulbs” surrounding axons. (B, Courtesy G. Richard Dickersin, MD, from *Diagnostic Electron Microscopy: A Text Atlas*, Igaku-Shoin Medical Publishers; 2000:984.)

polyneuropathies. Connective tissue disease–associated neuropathies are classically described as axonal polyneuropathies, but demyelination can also occur. These neuropathies are distinct from vasculitic peripheral neuropathies, which can arise as secondary manifestations of these same diseases.

Neuropathy Associated with Vasculitis. Vasculitis is a noninfectious and destructive inflammation of blood vessels that can involve and damage peripheral nerves with an axonal pattern. About one-third of patients with vasculitis have peripheral nerve involvement, and neuropathy may be the presenting feature. This often takes the form of mononeuritis multiplex, but mononeuropathy and polyneuropathy are also encountered.

The most common form of vasculitis associated with peripheral neuropathy is a localized form referred to as *nonsystemic vasculitic neuropathy*, which is not associated with any antineutrophil cytoplasmic antibodies (ANCA). The forms of systemic vasculitis (Chapter 11) that are most commonly associated with peripheral neuropathy are polyarteritis nodosa or anti-myeloperoxidase (MPO-) ANCA–associated vasculitides (microscopic polyangiitis and eosinophilic granulomatosis with polyangiitis [formerly Churg–Strauss syndrome]); association with antiproteinase-3 (PR3-) ANCA–associated granulomatosis with polyangiitis (formerly known as Wegener granulomatosis) is much less common. Regardless of the type of vasculitis, peripheral nerves involved by vasculitis typically show patchy axonal degeneration and loss, with some fascicles being more severely affected than others (i.e., asymmetric involvement). Perivascular inflammatory infiltrates are often present. Identification of blood vessels with characteristic forms of acute or chronic injury helps establish the diagnosis.

Infectious Neuropathies

Many infectious processes affect peripheral nerves. Among these, leprosy, diphtheria, and varicella-zoster cause relatively specific pathologic changes in nerves that are the focus here. These infectious diseases are discussed in more detail in Chapter 8.

Leprosy (Hansen Disease). Peripheral nerves are involved in both lepromatous and tuberculoid leprosy.

- In *lepromatous leprosy*, Schwann cells are invaded by *Mycobacterium leprae*, which proliferate and eventually infect other cells

including macrophages. Bacilli are readily identifiable with mycobacterial stains. There is early evidence of segmental demyelination and remyelination that progresses to axonal degeneration with loss of both myelinated and unmyelinated axons. As the infection advances, endoneurial fibrosis and multilayered thickening of the perineurial sheaths occur. Affected individuals develop a symmetric polyneuropathy that is most severe in the relatively cool distal extremities and in the face because lower temperatures favor *Mycobacterium leprae* growth. The infection prominently involves small pain fibers, and the resulting loss of sensation contributes to tissue damage, since the patient is rendered unaware of traumatic injuries. Thus, large ulcers may develop.

- **Tuberculoid leprosy** is characterized by an active cell-mediated immune response to *M. leprae* that usually manifests as dermal nodules containing granulomatous inflammation. The inflammation injures cutaneous nerves in the vicinity. Axons, Schwann cells, and myelin are lost, and there is fibrosis of the perineurium and endoneurium. In tuberculoid leprosy, affected individuals have much more localized nerve involvement. In contrast to lepromatous leprosy, bacilli are difficult to identify.

Lyme Disease. Lyme disease causes various neurologic manifestations in the second and third stages of the disease. These include polyradiculoneuropathy and unilateral or bilateral facial nerve palsies due to cranial neuropathy that are primarily axonal.

HIV/AIDS. Patients infected with human immunodeficiency virus (HIV) develop several patterns of peripheral neuropathy that are poorly understood, but all appear to be related in some way to immune dysregulation. Early-stage HIV infection may be associated with mononeuritis multiplex or demyelinating disorders that resemble Guillain-Barré syndrome or CIDP. More commonly, later stages of HIV infection are associated with a distal sensory neuropathy with an axonal pattern that is often painful.

Diphtheria. Peripheral nerve dysfunction results from the effects of diphtheria exotoxin. It produces an acute demyelinating peripheral neuropathy associated with prominent bulbar and respiratory muscle dysfunction, which can lead to death or long-term disability. The mechanism of action of diphtheria toxin is described in Chapter 8. Diphtheria is most commonly found in low-resource countries and is a continuing medical problem because of incomplete immunization or waning immunity in adults.

Varicella-Zoster Virus. Varicella-zoster is one of the most common viral infections of the peripheral nervous system. Following chickenpox, a latent infection persists within neurons of sensory ganglia. If the virus is reactivated, usually many years later, it may be transported along the sensory nerves to the skin. Here it infects keratinocytes, leading to a painful, vesicular skin eruption in a distribution that follows sensory dermatomes (*shingles*). Most common is the involvement of thoracic or trigeminal nerve dermatomes. The factors underlying reactivation of the virus are not fully understood, but decreased cell-mediated immunity is suspected to play a role. In a small proportion of patients, weakness is also apparent in the same distributions. Affected ganglia show neuronal death, usually accompanied by abundant mononuclear inflammatory cell infiltrates; focal necrosis and hemorrhage may also be found. Peripheral nerves show degeneration of axons belonging to the dead sensory neurons. Focal destruction of the large motor neurons of the anterior horns or cranial nerve motor nuclei may be seen at the corresponding levels. Intranuclear inclusions generally are not found in the peripheral nervous system.

Metabolic, Hormonal, and Nutritional Neuropathies

Diabetes. Diabetes is the most common cause of peripheral neuropathy. The prevalence of this complication depends on the duration of the disease; up to 50% of patients with diabetes overall and up to 80% of those who have had the disease for more than 15 years have evidence of peripheral neuropathy. Patients with type 1 and type 2 diabetes are affected (Chapter 24). Several distinct clinicopathologic patterns of diabetes-related peripheral neuropathy are recognized (described later), but the most common by far is an ascending distal symmetric sensorimotor polyneuropathy.

Pathogenesis

The mechanism of diabetic neuropathy is complex and not completely defined; both metabolic and secondary vascular changes are believed to contribute to the damage of axons and Schwann cells. Hyperglycemia causes the nonenzymatic glycosylation of proteins, lipids, and nucleic acids. The resulting advanced glycosylation end products (AGEs) may interfere with normal protein function and activate inflammatory signaling through the AGE-specific receptor. In a separate (polyol) pathway, excess glucose within cells is reduced to sorbitol, a process that depletes NADPH and increases intracellular osmolality. It may also activate signaling pathways through AGE receptors. These and other metabolic disturbances predispose peripheral nerves to injury by reactive oxygen species. In addition, the vascular injuries that occur in chronic diabetes due to hyperlipidemia and other metabolic alterations may cause ischemic damage of the nerves. Sensory axons are most involved followed by autonomic axons and less often motor axons. Sensory axons show a length-dependent dying-back pattern of injury resulting in a “stocking and glove” clinical presentation. Cell bodies are relatively preserved.

MORPHOLOGY

In individuals with a distal symmetric sensorimotor polyneuropathy, the predominant pathologic finding is an axonal neuropathy. Nerve biopsies show reduced numbers of axons. Variable degrees of ongoing axonal damage, marked by degenerating myelin sheaths and regenerative axonal clusters, may be present. Endoneurial arterioles show thickening, hyalinization, and intense periodic acid–Schiff positivity of their walls and extensive reduplication of basement membranes (Fig. 27.6). Extensive involvement and dropout of small, unmyelinated sensory nerve fibers (so-called C-fibers) found in the skin correlates with symptoms.

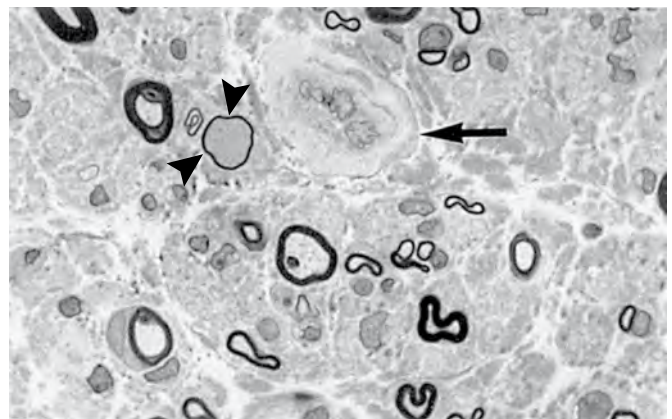


Fig. 27.6 Diabetic neuropathy. Light micrograph showing marked loss of myelinated fibers, a thinly myelinated fiber (arrowheads), and thickening of an endoneurial vessel wall (arrow).

Clinical Features

Distal symmetric diabetic polyneuropathy typically presents with sensory symptoms like numbness, loss of pain sensation, difficulty with balance, and paresthesias or dysesthesias in a stocking and glove distribution. Paresthesias or dysesthesias (abnormal painful sense of touch) are so-called “positive” symptoms—painful sensations that result from abnormal discharges of damaged nerves. Neuropathy leads to considerable morbidity, in particular an increased susceptibility to foot and ankle fractures and chronic skin ulcers, which may eventually lead to amputations.

Another manifestation of diabetes is dysfunction of the autonomic nervous system; this affects 20% to 40% of individuals with diabetes, nearly always in association with a distal sensorimotor polyneuropathy. Diabetic autonomic neuropathy has protean manifestations, including postural hypotension, incomplete emptying of the bladder (resulting in recurrent infections), and sexual dysfunction.

Some affected individuals, especially older adults with a long history of diabetes, develop a peripheral neuropathy that manifests with asymmetric presentations, including *mononeuropathy*, *cranial neuropathy*, and *radiculoplexus neuropathy*. The last-mentioned is a painful acute disorder that presents in the distribution of the brachial or lumbosacral nerve plexus. It is often monophasic and can improve over several months. These asymmetric manifestations may be caused by microvascular disease.

Other Metabolic, Hormonal, and Nutritional Neuropathies. A diverse group of metabolic, hormonal, and nutritional disorders are associated with peripheral neuropathy, including:

- **Uremic neuropathy.** Most individuals with renal failure develop peripheral neuropathy. Typically, this is a distal symmetric neuropathy that may be asymptomatic or associated with muscle cramps, distal dysesthesias, and diminished deep tendon reflexes. In these patients, axonal degeneration is the primary event; occasionally there is secondary demyelination. Regeneration and recovery are common after dialysis.
- **Thyroid dysfunction.** Hypothyroidism can lead to compression mononeuropathies such as carpal tunnel syndrome or cause a distal symmetric, predominantly sensory, axonal polyneuropathy. In rare cases, hyperthyroidism is associated with a neuropathy resembling Guillain-Barré syndrome but it is more commonly associated with a myopathy.
- **Vitamin B₁₂ (cyanocobalamin) deficiency** classically results in subacute combined degeneration with damage to long tracts in the spinal cord (Chapter 28) and peripheral nerves.
- **Deficiencies of vitamin B₁ (thiamine), vitamin B₆ (pyridoxine), vitamin E (tocopherol), folate, copper, and zinc** all have been associated with peripheral neuropathy.

Toxic Neuropathies

Peripheral neuropathies may appear after exposure to industrial or environmental chemicals, biologic toxins, or therapeutic drugs. Important causes of toxic peripheral nerve damage include alcohol (independent of associated nutritional deficiencies), heavy metals (lead, mercury, arsenic, and thallium), and organic solvents. Various medications can cause toxic nerve damage, but the most notorious are chemotherapeutic agents. These include vinca alkaloids and taxanes, microtubule inhibitors that interfere with axonal transport, and cisplatin, which may cause a neuropathy. In addition, immune checkpoint inhibitors, a new class of cancer immunotherapy, can cause peripheral neuropathy with multiple phenotypes including non-length-dependent polyradiculopathy with axonal or demyelinating features, cranial neuropathy, and sensory neuronopathy.

Neuropathies Associated With Malignancy

Neuropathies associated with cancers may stem from local effects, complications of therapy, paraneoplastic effects, or (in the case of B-cell tumors) tumor-derived immunoglobulins.

- **Direct infiltration or compression of peripheral nerves** by tumors is a common cause of axonal mononeuropathy and may be a presenting symptom of cancer. These neuropathies include brachial plexopathy from neoplasms of the apex of the lung, obturator palsy from pelvic malignant neoplasms, and cranial nerve palsies from intracranial tumors or tumors of the base of the skull. A polyradiculopathy involving the lower extremity may develop when the cauda equina is involved by meningeal carcinomatosis. Neurolymphomatosis occurs when tumor cells infiltrate peripheral nerves and is most often caused by non-Hodgkin B-cell lymphoma (Chapter 13). In approximately half of the patients with neurolymphomatosis, there is no known systemic lymphoma at the time of diagnosis.
- **Paraneoplastic neuropathies.** These can occur at any time during the patient's course, but often precede the diagnosis of the underlying tumor. Sensorimotor axonal neuropathy is the most common paraneoplastic form, but a pure sensory neuronopathy, chronic inflammatory demyelinating polyneuropathy, plexopathy, and autonomic neuropathy may also be seen. Paraneoplastic sensory neuronopathy is most commonly associated with small cell lung cancer. Antibodies that recognize proteins expressed by cancer cells and normal neurons (for example, anti-Hu antibodies, also known as antineuronal nuclear antibody-type 1, ANNA-1) are often present, but the damage appears to be mediated by a CD8+ cytotoxic T-cell attack on dorsal root ganglion cells. Sensory symptoms usually start distally in an asymmetric and multifocal pattern. Other patients with anti-CV2 autoantibodies (which recognize CRMP5, an intracellular signaling protein) tend to present with a painful axonal asymmetric sensorimotor neuropathy.
- **Neuropathies associated with monoclonal gammopathies.** Neoplastic B cells may secrete monoclonal immunoglobulins or immunoglobulin fragments (so-called paraproteins) that damage nerves. For example, tumors that secrete IgM immunoglobulin may be associated with a demyelinating peripheral neuropathy. In most cases, the pathogenic IgM paraprotein is thought to bind directly to myelin-associated antigens such as myelin-associated glycoprotein. Deposition of IgM can be seen ultrastructurally between the membrane layers of the myelin sheath. IgG or IgA paraproteins may also be associated with peripheral neuropathy. One distinctive presentation is **POEMS syndrome** (polyneuropathy, organomegaly, endocrinopathy, monoclonal gammopathy, and skin changes), in which patients develop a demyelinating neuropathy associated with deposition of paraprotein between noncompacted myelin lamellae. Finally, excess immunoglobulin light chain may deposit as amyloid (Chapter 6), which can lead to an axonal peripheral neuropathy due to vascular insufficiency or a direct toxic effect.

Neuropathies Caused by Physical Forces

Peripheral nerves are commonly injured by trauma or entrapment that leads to an axonal degeneration.

- **Lacerations** result from cutting injuries and from sharp fragments of fractured bone, both of which may sever a nerve.
- **Avulsion** of a nerve may occur when tension is applied, often to one of the limbs.
- **Compression neuropathy (entrapment neuropathy)** occurs when a peripheral nerve is chronically subjected to increased pressure, often within an anatomic compartment. Carpal tunnel syndrome, the most common entrapment neuropathy, results from compression of the median nerve at the level of the wrist within the compartment delimited by the transverse carpal ligament. Females are more commonly

affected than males, and the problem is frequently bilateral. The disorder may be observed in association with many conditions, including tissue edema, pregnancy, inflammatory arthritis, hypothyroidism, amyloidosis (especially related to β_2 -microglobulin deposition in individuals on renal dialysis), acromegaly, diabetes mellitus, and excessive repetitive motions of the wrist. Symptoms are limited to dysfunction of the median nerve and typically include numbness and paresthesias of the tips of the thumb and first two digits. Other nerves prone to compression neuropathies include the ulnar nerve at the level of the elbow, the peroneal nerve at the level of the knee, and the radial nerve in the upper arm; the latter occurs from sleeping with the arm in an awkward position (“Saturday night palsy”). Another form of compression neuropathy is found in the foot, affecting the interdigital nerve at intermetatarsal sites. This problem, which occurs more often in females than in males, leads to foot pain (metatarsalgia) and is associated with a histologic lesion called a *Morton neuroma*, which is marked by significant perineural fibrosis.

Inherited Peripheral Neuropathies

Inherited peripheral neuropathies are a group of genetic disorders with overlapping clinical phenotypes that often present in adults. Because of the delayed onset and the possibility of de novo mutations without a family history, genetically determined neuropathy must be considered in the differential diagnosis for any patient who presents with peripheral neuropathy. The major types include (1) hereditary motor and sensory neuropathies, commonly known as *Charcot-Marie-Tooth* (CMT) disease, described next, a genetically heterogeneous disorder caused by mutations in genes that encode proteins involved in the structure and function of the peripheral nerve axon or the myelin sheath; (2) hereditary motor neuropathies; (3) hereditary sensory neuropathies, with or without autonomic neuropathy; and (4) other inherited conditions causing neuropathy, including familial amyloidosis and inherited metabolic diseases.

Historically, these diseases were classified based on their inheritance pattern and clinical features. Now there is a continuously growing list of over 100 altered genes that are linked to these diseases. The complexity of the genetics of inherited neuropathies is a reflection of the complicated homeostatic mechanisms that sustain normal peripheral nerve function. There is no simple unifying concept tying all the implicated genes together; some of the involved genes are illustrated in Fig. 27.1. Some of the more common and distinctive types of inherited peripheral neuropathies are described in brief.

Hereditary Motor and Sensory Neuropathies (Charcot-Marie-Tooth Disease). These are by far the most common inherited peripheral neuropathies, affecting up to 1 in 2500 people. The initial description of these disorders, based on clinical features, was deceptively simple—an inherited disease associated with distal muscle atrophy, sensory loss, and foot deformities. It is now appreciated that this phenotype can result from mutations in more than 50 different genes. Current classification systems distinguish hereditary motor and sensory neuropathies based on the mode of inheritance and the pattern of injury (e.g., axonal or demyelinating). Demyelinating forms of Charcot-Marie-Tooth (CMT) disease are associated with morphologic features of demyelination and remyelination including Schwann cell hyperplasia and onion bulb formation, which may be so profound that the involved nerve is palpably enlarged. Listed are the most common types, which together account for over 90% of patients with CMT disease:

- *CMT1 encompasses a group of autosomal dominant demyelinating neuropathies* that collectively are the most common subtype of hereditary motor and sensory neuropathy. *CMT1A* accounts for about 55% of genetically defined CMT. It is caused by a duplication of a

region on chromosome 17 that includes the peripheral myelin protein 22 (*PMP22*) gene. The disease usually presents in the first or early second decade of life as a slowly progressive distal demyelinating motor and sensory neuropathy with motor symptoms predominating. *CMT1B* is a demyelinating neuropathy caused by pathogenic mutations in the myelin protein zero (*MPZ*) gene that result in overexpression of this major structural protein. It accounts for about 9% of genetically defined cases of CMT. *CMT1B* has a highly variable timing of presentation, with onset in infancy, childhood, or adulthood.

- *CMTX* encompasses X-linked forms of CMT disease. *CMT1X* is the most common of these, accounting for 15% of genetically defined cases of CMT. It is linked to mutations in the *GJB1* gene, which encodes connexin32, also known as gap junction protein beta 1, a gap junction component that is expressed in Schwann cells. Males are typically more severely affected, but female patients can also manifest disease with a range in severity due to variable X-inactivation.
- *CMT2* includes autosomal dominant neuropathies associated with axonal rather than demyelinating injury. *CMT2A* is the most common subtype, accounting for 4% of all CMT disease. It is caused by mutations in the *MFN2* gene, which encodes mitofusin 2, required for normal mitochondrial fusion. The phenotype is typically severe, with disease onset in early childhood and sensory symptoms predominating.

Hereditary Sensory Neuropathies With or Without Autonomic Neuropathy. This is a diverse group of diseases marked by loss of sensation and variable autonomic disturbances, but with preserved motor strength. Loss of pain and temperature sensation is the most common symptom. The inability to sense pain leads to traumatic injury of the hands and feet. These are typically axonal neuropathies.

Familial Amyloid Polyneuropathies. These are hereditary disorders characterized by amyloid deposition within peripheral nerves. Most are caused by germline mutations of the transthyretin (*TTR*) gene. The mutated transthyretin protein, which is normally involved in binding and transport of thyroid hormone, is prone to deposit as amyloid fibrils in a number of tissues including peripheral nerve (Chapter 6). The clinical presentation is similar to that of hereditary sensory and autonomic neuropathies. Amyloid can be identified histologically by Congo red staining viewed under polarized light and/or by characteristic findings on electron microscopy.

KEY CONCEPTS

Peripheral Neuropathies

- Anatomic patterns include mononeuropathy, mononeuritis multiplex, polyneuropathy, and polyradiculoneuropathy.
- Damage may occur primarily in Schwann cells (demyelinating neuropathy), axons (axonal neuropathy), or central neurons (neuronopathy); mixed patterns of injury occur.
- Inflammatory disease, infections, metabolic abnormalities, toxic injury, trauma, paraneoplastic disorders, and genetic defects all can cause peripheral neuropathy.
- Diabetes mellitus is the most common cause of peripheral neuropathy, which most often presents as a distal symmetric axonal polyneuropathy.
- Guillain-Barré syndrome and chronic inflammatory demyelinating polyradiculoneuropathy are the major acute and chronic acquired demyelinating peripheral neuropathies, respectively.
- Inherited peripheral neuropathies are genetically and phenotypically diverse disorders that often present in adulthood and may be marked by sensory, motor, or autonomic dysfunction, alone or in combination.

DISEASES OF THE NEUROMUSCULAR JUNCTION

The neuromuscular junction is a specialized structure located at the interface of motor nerve axons and skeletal muscle that serves to control muscle contraction. Neuromuscular junctions are found midway along the length of myofibers. Here, the distal end of each motor axon branches into smaller processes that terminate in bulbous synaptic swellings (boutons) on the myofibers innervated by the respective motor unit. Upon depolarization, these presynaptic nerve terminals release acetylcholine (ACh) into the synaptic cleft, the space separating the nerve endings from the myofiber membrane (referred to as the *sarcolemma*). The postsynaptic sarcolemma is characterized by complex infoldings and exhibits distinct specializations including localized clustering of ACh receptors. These receptors are responsible for the initiation of signals leading to muscle contraction.

Regardless of cause, patients with disorders that impair the function of neuromuscular junctions complain of painless weakness and fatigue. Autoantibodies that recognize key neuromuscular junction proteins are the most common cause of disrupted neuromuscular transmission, as found in *myasthenia gravis* (literally, “severe weakness”). Understandably, inherited defects in specialized neuromuscular junction proteins are also associated with myasthenic syndromes. Disorders caused by toxins that alter neuromuscular transmission are rarely encountered but they held an important role historically in elucidating how the neuromuscular junction functions.

Antibody-Mediated Diseases of the Neuromuscular Junction

Myasthenia Gravis

Myasthenia gravis is an autoimmune disease that is usually associated with autoantibodies directed against ACh receptors. It has a prevalence of 150 to 200 per 1 million and shows a bimodal age distribution. In young adults, the female-to-male ratio is 2:1, but in older adults there is a male predominance.

Pathogenesis

About 85% of patients have autoantibodies against postsynaptic ACh receptors, while most of the remaining patients have antibodies against the sarcolemmal protein muscle-specific receptor tyrosine kinase (MuSK). These autoantibodies are pathogenic, as the disease can be transferred to animals with serum from affected individuals, and therapeutic maneuvers that decrease autoantibody levels are associated with a reduction in symptoms.

The mechanism of action of the various autoantibodies appears to differ. Anti-ACh receptor antibodies are thought to lead to the aggregation and degradation of the receptors, as well as to damage of the postsynaptic membrane through complement fixation. As a result, postsynaptic membranes show alterations in morphology and are depleted of ACh receptors. This limits the ability of myofibers to respond to ACh. Autoantibodies directed against muscle-specific receptor tyrosine kinase do not fix complement. Instead, these antibodies seem to interfere with the trafficking and clustering of ACh receptors within the sarcolemmal membrane, the net effect again being decreased ACh receptor function.

There is a strong association between pathogenic anti-ACh receptor autoantibodies and thymic abnormalities. Approximately 10% of patients with myasthenia gravis have a *thymoma*, a tumor of thymic epithelial cells (Chapter 13). An additional 30% of patients (and particularly young patients) have thymic hyperplasia, which is accompanied by the appearance of B-cell follicles in the thymus. The thymus normally contains cells that express peripheral tissue

antigens, including skeletal muscle proteins. It is hypothesized that both thymoma and thymic hyperplasia disrupt normal expression of these tissue antigens, leading to failure of self-tolerance and autoimmunity against ACh receptors expressed in thymic cells. In contrast, thymic abnormalities are usually absent in cases of myasthenia gravis that occur in older patients; the basis for the development of autoantibodies in such cases is unknown.

Clinical Features

Patients with anti-ACh receptor antibodies typically present with fluctuating weakness that worsens with exertion and often over the course of the day. Diplopia and ptosis due to involvement of extraocular muscles are common and distinguish myasthenia gravis from most myopathies, in which involvement of extraocular muscles is unusual. In some patients, symptoms are confined to ocular muscles, while others develop generalized weakness that can be so severe as to require mechanical ventilation. Cases with antibodies against muscle-specific receptor tyrosine kinase differ from typical cases by exhibiting more focal muscle involvement, particularly of neck, shoulder, facial, respiratory, and bulbar muscles.

Diagnosis is based on clinical history, physical findings, identification of autoantibodies, and electrophysiologic studies. Electrophysiologic studies reveal a *decrease in muscle response with repeated stimulation*, a characteristic of this disorder. Overall mortality has dropped from over 30% in the 1950s to less than 5% with current therapies. Acetylcholinesterase inhibitors that increase the half-life of ACh are the first line of treatment. Other treatments, such as plasmapheresis, immunosuppressive medications (e.g., glucocorticoids, cyclosporine, rituximab), and a new medication that increases catabolism of IgG can bring symptoms under control by decreasing autoantibody levels. Other new therapies target the complement pathway, lending further support to the importance of complement-mediated damage in the pathogenesis of myasthenia gravis. Thymectomy is often effective in patients with thymoma but is of uncertain benefit in those with thymic hyperplasia or lacking thymic abnormalities.

Lambert-Eaton Myasthenic Syndrome

Lambert-Eaton myasthenic syndrome is an autoimmune disorder caused by antibodies that block ACh release by inhibiting a presynaptic calcium channel. In contrast to myasthenia gravis, *rapid repetitive stimulation increases muscle response*. Muscle strength is augmented after a few seconds of muscle activity. Patients typically present with weakness of their extremities. In about half of cases there is an underlying malignancy, most often small cell carcinoma of the lung. Symptoms may precede the diagnosis of cancer, sometimes by years. It is thought that the stimulus for autoantibody formation in paraneoplastic cases may be the expression of the same calcium channel in the neoplastic cells. Patients without cancer often have other autoimmune diseases, such as vitiligo or thyroid disease. Treatment consists of drugs that increase ACh release by depolarizing synaptic membranes and immunosuppressive agents, such as those used to treat myasthenia gravis.

Congenital Myasthenic Syndromes

These rare disorders, which most commonly have an autosomal recessive mode of inheritance, are marked by varying degrees of muscle weakness. Causative mutations have been identified in genes encoding several different presynaptic, synaptic, and postsynaptic proteins. The most common of these are loss-of-function mutations in the gene encoding the ϵ -subunit of the ACh receptor. Another group of mutations affect proteins that are important in normal clustering of ACh receptors on postsynaptic membranes. Many patients with congenital myasthenic syndromes present in the perinatal period with poor muscle tone, external eye muscle weakness, and breathing difficulties, but

others have milder forms of the disease and may not come to clinical attention until adolescence or adulthood. The clinical presentation, response to drugs such as acetylcholinesterase inhibitors, and prognosis depend largely on the underlying mutation.

Disorders Caused by Toxins

Botulism is caused by exposure to a neurotoxin (botulinum toxin, popularly known as Botox) that is produced by the anaerobic gram-positive organism *Clostridioides botulinum*. Botox acts by blocking the release of ACh from presynaptic neurons (Chapter 8). Curare is the common name for plant-derived muscle relaxants that block ACh receptors, resulting in flaccid paralysis. It was initially discovered and used as poison on arrow tips by indigenous people in the Amazon rain forest. At one time it was used as a muscle relaxant during certain forms of surgery but has now been supplanted by other drugs with a similar mechanism of action.

KEY CONCEPTS

Diseases of the Neuromuscular Junction

- Disorders of neuromuscular junctions present with painless weakness.
- Myasthenia gravis and Lambert-Eaton myasthenic syndrome, the most common forms, are both immune mediated, being caused by antibodies to postsynaptic membrane proteins (ACh receptor or muscle-specific receptor tyrosine kinase) and presynaptic calcium channels, respectively.
- Myasthenia gravis is often associated with thymic hyperplasia or thymoma, frequently involves ocular muscles, and is marked by fluctuating weakness that worsens with exertion.
- Lambert-Eaton myasthenic syndrome presents with weakness in the extremities that improves with repetitive stimulation and is often a paraneoplastic disorder associated with lung cancer.
- Genetic defects in neuromuscular junction proteins give rise to congenital myasthenic syndromes.
- Biologic toxins such as botulinum toxin can block neuromuscular transmission by blocking the release of ACh from presynaptic neurons.

DISEASES OF SKELETAL MUSCLE

Skeletal muscle has unique structural, cellular, and molecular characteristics and accordingly unique patterns of injury and repair. During embryogenesis, skeletal muscle develops through the fusion of mononucleated precursor cells (*myoblasts*) into multinucleated myotubes. These subsequently mature into *myofibers* (muscle fibers) of varying length that contain thousands of nuclei. In adult tissues, myofibers are organized into fascicles, each associated with a small pool of tissue stem cells referred to as satellite cells, which can contribute to muscle regeneration following injury (outlined later). Myofibers are of two main functional types, type I and type II (Table 27.2), which are admixed in a “checkerboard” pattern in normal skeletal muscle.

Skeletal Muscle Atrophy

Skeletal muscle atrophy is a common feature of many disorders. Loss of innervation, disuse, cachexia, old age, and primary myopathies all can produce muscle atrophy and loss of muscle mass. Although muscle atrophy may occur under many conditions, certain patterns of atrophy suggest specific underlying etiologies, as illustrated by the following examples:

- *Clusters or groups of atrophic fibers* are seen in neurogenic disease.
- *Perifascicular atrophy* is seen in dermatomyositis (see later).
- *Type II fiber atrophy* with sparing of type I fibers is seen with prolonged corticosteroid therapy or disuse.

TABLE 27.2 Muscle Fiber Types

	Type I	Type II
Action	Sustained force	Fast movement
Activity type	Aerobic exercise	Anaerobic exercise
Power produced	Low	High
Resistance to fatigue	High	Low
Lipid content	High	Low
Glycogen content	Low	High
Energy metabolism	Low glycolytic capacity, high oxidative capacity	High glycolytic capacity, low oxidative capacity
Mitochondrial density	High	Low
Enzyme activity	NADH-TR, dark staining ATPase at pH 4.3, dark staining ATPase at pH 9.4, light staining	NADH-TR, light staining ATPase at pH 4.3, light staining ATPase at pH 9.4, dark staining
Myosin heavy chain gene expressed	<i>MYH7</i>	<i>MYH2</i> , <i>MYH4</i> , <i>MYH1</i>
Color	Red (high myoglobin content)	Pale red/tan (low myoglobin content)
Prototype	Soleus	Pectoral

ATPase, Adenosine triphosphatase; NADH-TR, nicotinamide adenine dinucleotide, reduced form, tetrazolium reductase.

Neurogenic and Myopathic Changes in Skeletal Muscle

Weakness and disruption of skeletal muscle function may be caused by direct injury to myofibers (myopathic injury) or by loss of muscle innervation (neurogenic injury). Each type is described next.

Neurogenic Changes

Neurogenic changes are associated with *fiber type grouping* and *grouped atrophy* (Fig. 27.7), both of which stem from the disruption of muscle innervation. The key to understanding these abnormalities is to recognize that muscle fiber type is determined by the innervating motor neuron. An individual motor neuron, its axon, and the myofibers innervated by it form a *motor unit* and all myofibers of a motor unit are therefore of the same type. Following denervation, myofibers undergo atrophy, often assuming a flattened, angulated shape. Reinnervation restores fiber size and shape but may make a denervated myofiber part of a different motor unit, in turn leading to a potential change in fiber type. In the face of ongoing axonal or neuronal damage and dropout, residual motor axons may innervate increasingly larger numbers of myofibers, leading to enlargement of motor units, each comprised of a single type of muscle fiber (fiber type grouping). These large motor units are also susceptible to grouped atrophy if the innervating axon is damaged. Neurogenic atrophy is a common finding in processes like amyotrophic lateral sclerosis (ALS, Chapter 28), radiculopathy from degenerative spine disease, or peripheral neuropathy.

Spinal muscular atrophy is a neuropathic disorder in which loss of motor neurons leads to muscle weakness and atrophy. Spinal muscular atrophy is an autosomal recessive disorder with an incidence of 1 in 6000 births and is caused by loss-of-function mutations in the *SMN1* (survival of motor neuron-1) gene (Chapter 28). *SMN1* regulates the homeostasis of motor neurons through several processes including a role in the spliceosome and biogenesis of ribonucleoproteins.

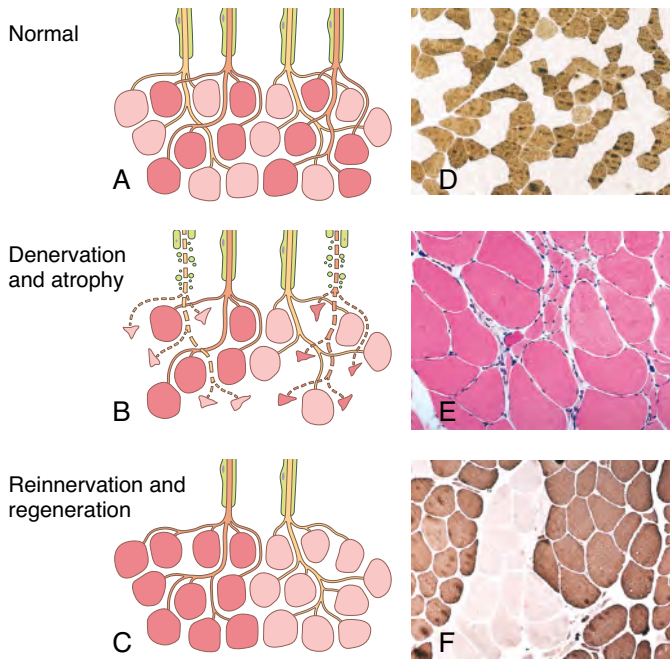


Fig. 27.7 Muscle atrophy. (A) This diagrammatic representation of four normal motor units shows a normal checkerboard-type admixture of light- and dark-stained fibers of opposite type. (B) Damage to innervating axons leads to loss of trophic input and atrophy of myofibers. (C) Reinnervation of myofibers can lead to a switch in fiber type and segregation of fibers of like type. As illustrated here, reinnervation is also often associated with an increase in motor unit size, with more myofibers innervated by an individual axon. (D) Normal muscle has a checkerboard-type distribution of type I (light) and type II (dark) fibers on this adenosine triphosphatase reaction (pH 9.4) corresponding to the schematic representation in A. (E) Clustered flattened, “angulated” atrophic fibers (*group atrophy*) are a typical finding associated with disrupted innervation. (F) With ongoing denervation and reinnervation, large clusters of fibers appear that all share the same fiber type (*type grouping*), corresponding to C.

Deficiency of SMN1 leads to loss of motor neurons that may start in utero. The resulting denervation or the lack of normal initial innervation of skeletal muscle may lead to characteristic morphologic changes. In the severe infantile form, large zones of severely atrophic, round myofibers are admixed with scattered normal-sized or hypertrophied myofibers, found individually or in small groups (Fig. 27.8). These normal or hypertrophied fibers are those that gained innervation from remaining motor neurons.

Myopathic Changes

In contrast to neurogenic changes, most primary myopathies are associated with a distinct set of morphologic changes that include:

- **Segmental myofiber degeneration and regeneration** are seen when only a segment of a tube-shaped myofiber undergoes necrosis. Degeneration is associated with release of cytoplasmic enzymes such as creatine kinase (CK) into the blood, making these useful clinical markers of muscle damage. The sarcomeres and other components of the damaged myofiber segment are engulfed by macrophages in a process termed *myophagocytosis*. Removal of the degenerated debris sets the stage for regeneration. Fusion of activated satellite cells to damaged myofibers is an important step for regeneration. Eventually, new sarcomeres are generated, and the continuity of the original myofiber is restored. Regenerating myofibers are rich in

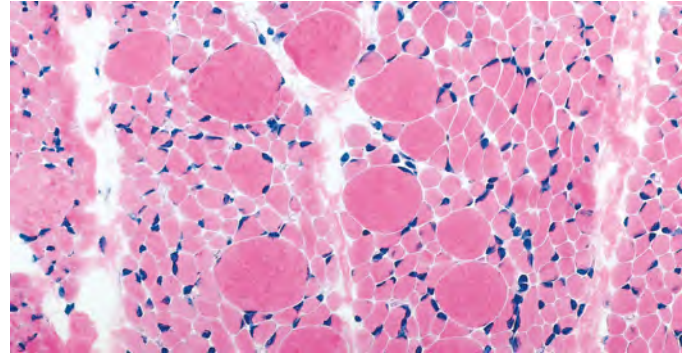


Fig. 27.8 Spinal muscular atrophy with only rare hypertrophied myofibers admixed with numerous atrophic rounded myofibers. The larger fibers are those that are innervated and have undergone compensatory hypertrophy.

RNA and therefore blue (basophilic) in hematoxylin and eosin-stained sections. They have enlarged nuclei with prominent nucleoli that are often randomly distributed in the cytoplasm, instead of occupying their normal subsarcolemmal location. Depending on the nature of the primary insult, atrophic myofibers may also be seen. Regeneration can restore normal muscle following an acute, transient injury, but in chronic disease states, regeneration often fails to keep pace with damage. In this setting, muscles often show dropout of myofibers, endomysial fibrosis (collagen deposition), and fatty replacement.

- **Myofiber hypertrophy** can be seen as a physiologic adaptation to exercise or in association with certain chronic myopathic conditions.
- **Cytoplasmic inclusions** in the form of vacuoles, protein aggregates, or clustered organelles are characteristic of several primary forms of myopathy.

Inflammatory Myopathies

Muscle injury secondary to inflammation can be seen following infections such as trichinosis (Chapter 8) but is most often caused by an autoimmune process. These autoimmune diseases include systemic connective tissue diseases such as systemic lupus erythematosus and systemic sclerosis (Chapter 6), as well as sarcoidosis. In other, idiopathic inflammatory myopathies, muscle involvement is the main manifestation of the disease. Historically, these inflammatory myopathies have been classified into three main types, dermatomyositis, polymyositis, and inclusion body myositis, but data on associated antibodies and clinical features has led to a more detailed and granular view of these diseases, which are now classified into the following: (1) *Dermatomyositis* with distinct subtypes; (2) *Anti-synthetase syndrome*; (3) *Immune-mediated necrotizing myopathy*; (4) *Sporadic inclusion body myositis*; (5) *Overlap myositis*; and (6) *Polymyositis*. Most patients that were diagnosed with polymyositis in the past are now included in one of the other idiopathic inflammatory myopathies. The group of overlap myositis includes patients with systemic connective tissue diseases that develop skeletal muscle involvement, mentioned above. The remaining five groups are discussed next.

Dermatomyositis

Dermatomyositis (DM) is a systemic autoimmune disease that typically includes proximal muscle weakness and skin changes as key clinical features.

Pathogenesis

Dermatomyositis includes a group of inflammatory myopathies that are characterized by dermatomyositis-specific antibodies (DMSAs) and prominent type I interferon (IFN)-signaling pathway activation. The DMSAs are associated with distinctive clinical and histopathologic features, discussed later. The exact role of these antibodies in the pathogenesis of the disease process is unknown. The activation of type I IFN signaling is a consistent feature of dermatomyositis and some systemic autoimmune diseases, notably lupus (Chapter 6). The presence of tubuloreticular inclusions in capillary endothelial cells and the overexpression of myxovirus resistant protein A (MxA), an IFN-induced antiviral protein, are viewed as histopathologic markers of abnormal IFN signaling, but the role of IFN in disease pathogenesis is not established. Vasculopathic changes are common, including *telangiectasias* (dilated capillary loops) in the nail folds, eyelids, and gums and dropout of capillary vessels in skeletal muscle, that are responsible for at least some of the manifestations of dermatomyositis. It is postulated that the accumulation of IFN-induced proteins in muscle and vascular cells has a toxic effect that is responsible for at least some of the injury.

MORPHOLOGY

The classic feature of dermatomyositis is **perifascicular atrophy**, a term that describes the preferential atrophy of fibers at the edge of muscle fascicles (Fig. 27.9B). However, it is found in only about 50% of cases and is especially associated with anti-TIF- γ as well as anti-Mi-2 antibodies. Other common findings include perimysial perivascular inflammatory cell infiltrates rich in T lymphocytes, perifascicular necrosis of myofibers (especially in cases with anti-Mi-2 antibodies), and microinfarcts (especially in cases with anti-NXP-2 antibodies). Deposition of C5b-9 (the membrane attack complex of complement) in capillaries in perifascicular areas is seen in a subset of cases and thought to underlie dropout of capillaries. MxA expression in muscle cells and the formation of **tubuloreticular endothelial inclusions** on electron microscopic studies are, as mentioned, a reflection of the IFN response.

Clinical Features

Muscle weakness is subacute in onset, symmetric, and often accompanied by myalgias. It typically affects the proximal muscles first. As a result, tasks such as getting up from a chair and climbing steps become increasingly difficult. Fine movements controlled by distal muscles are affected only late in the disease. Associated myopathic changes on electrophysiologic studies and elevation in serum creatine kinase levels are reflective of muscle damage. Various skin manifestations are seen in dermatomyositis, but the most characteristic ones are a lilac-colored discoloration of the upper eyelids associated with periorbital edema (*heliotrope rash*) (Fig. 27.9A) and a scaling erythematous eruption or dusky red patches over the knuckles, elbows, and knees (*Gottron papules*). Dysphagia resulting from involvement of oropharyngeal and esophageal muscles occurs in one-third of the affected individuals.

The clinical features vary in individuals with different antibody profiles.

- Antibodies against *TIF1- γ* (a signaling protein): These patients often have prominent skin involvement, dysphagia, and a significant association with a diagnosis of cancer in those over the age of 38 years.
- Antibodies against *Mi-2* (a nucleosome protein): These patients usually have high serum CK levels because of severe muscle involvement with prominent necrosis.
- Antibodies against *MDA5* (an IFN-induced cytosolic viral sensor): These patients often have little or no muscle involvement clinically and only mild changes on muscle biopsy. They fall into different subgroups. Some of these patients have rapidly progressive interstitial lung disease associated with a poor prognosis, while others have skin manifestations combined with joint involvement causing arthralgias and a comparably favorable prognosis.
- Antibodies against *NXP-2* (a nuclear matrix protein): These patients typically have weakness related to muscle involvement but limited cutaneous involvement. Some develop dystrophic calcifications in superficial soft tissues referred to as calcinosis.

Dermatomyositis is the most common inflammatory myopathy encountered in children. Juvenile patients are enriched in the group with anti-NXP-2 antibodies.

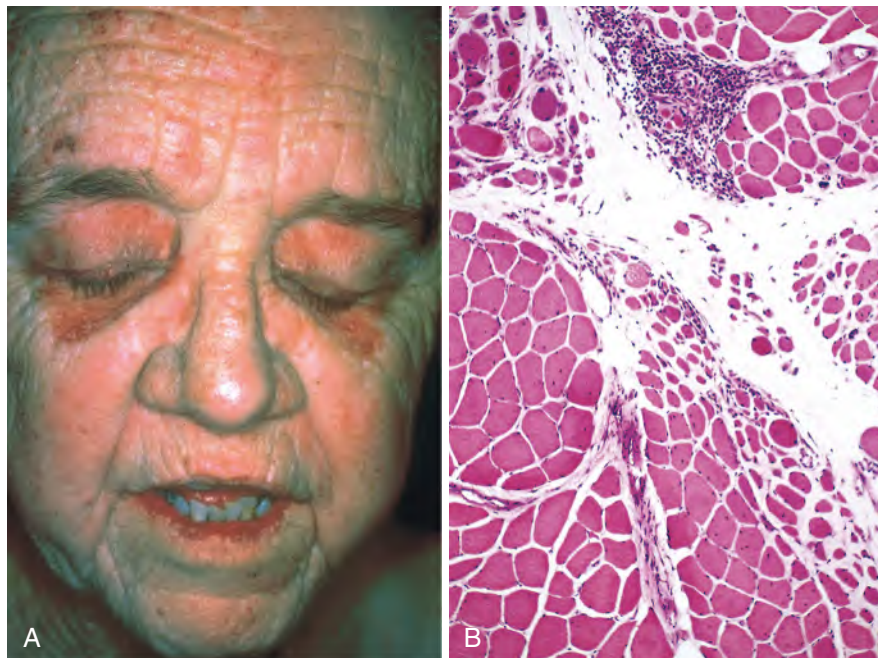


Fig. 27.9 Dermatomyositis. (A) Heliotrope rash affecting the eyelids. (B) The histologic appearance of muscle shows perifascicular atrophy of muscle fibers and inflammation. (Courtesy Dr. Dennis Burns, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Antisynthetase Syndrome

Antisynthetase syndrome is a disease that involves muscle and other tissues and is defined by the presence of serum antibodies directed against aminoacyl transfer RNA synthetase. The most common and prototypical form is associated with anti-Jo-1 antibodies but cases with other antisynthetase antibodies are described including anti-PL-7, anti-PL-12, anti-EJ, and anti-Ha. Clinical manifestations of antisynthetase syndrome include muscle disease (myositis with muscle weakness and myalgias), interstitial lung disease, and arthritis. Other rare features may include arthralgias, fever, Raynaud phenomenon, and skin manifestations. The latter include changes described as “mechanic’s hands,” hyperkeratotic, fissuring patches on the radial aspects of the fingers and the ulnar aspect of the thumb. The histopathologic features of the myositis often include perifascicular atrophy and necrosis that can mimic dermatomyositis but other cases may show more random distribution of necrotic myofibers.

Immune-Mediated Necrotizing Myopathy

Immune-mediated necrotizing myopathy (IMNM) is characterized by prominent muscle necrosis with high CK levels and rapidly progressive weakness. Associated systemic manifestations like skin changes, arthritis, and interstitial lung disease are seen only in a small subset of patients. IMNM is associated with antibodies against the ribonucleoprotein SRP or against-HMG-CoA reductase, an enzyme involved in cholesterol synthesis. Patients with anti-SRP antibodies usually have more severe disease, more often have dysphagia, and may develop cardiac involvement. Muscle biopsies typically show prominent myofiber damage with randomly admixed necrotic and regenerating myofibers and a paucity of associated inflammatory infiltrates. Anti-HMG-CoA reductase antibodies can be seen in patients with prior exposure to statins but are also reported in statin-naïve patients.

Sporadic Inclusion Body Myositis

Sporadic inclusion body myositis (sIBM) is a disease of late adulthood that typically affects patients older than 50 years and is the most common inflammatory myopathy in patients older than 65 years. Most affected individuals present with slowly progressive muscle weakness that tends to be most severe in the quadriceps and the distal upper extremity muscles. Dysphagia from esophageal and pharyngeal muscle involvement is not uncommon. Laboratory studies usually show relatively mild creatine kinase level elevations reflecting a more smoldering disease course. Antibodies to cytosolic 5'-nucleotidase 1A (cN1A) have been described in association with many cases of sIBM.

MORPHOLOGY

Sporadic inclusion body myositis is associated with a distinct set of biopsy changes in typical and established cases. These include:

- Patchy, often endomysial mononuclear inflammatory cell infiltrates rich in CD8+ T cells
- Focal invasion of normal appearing myofibers by inflammatory cells
- Increased sarcolemmal expression of major histocompatibility complex class I antigens
- Abnormal cytoplasmic inclusions described as “rimmed vacuoles” (Fig. 27.10) that contain abnormal lysosomal aggregates and protein aggregates typically associated with neurodegenerative diseases, like beta-amyloid, TDP-43, and ubiquitin
- On electron microscopy, distinctive filamentous aggregates often found in abnormal lysosomal structures
- Endomysial fibrosis and fatty replacement, reflective of a chronic disease course

Whether inclusion body myositis is indeed an inflammatory condition or a degenerative process with secondary inflammatory changes remains an unresolved question. It has certain features in common with other idiopathic inflammatory myopathies but also shares some features with neurodegenerative diseases, such as the presence of abnormal protein aggregates in muscle cells. Furthermore, there are several familial inclusion body myopathies that are also associated with chronic myopathic changes and rimmed vacuoles. These usually lack any associated inflammation—hence the designation *inclusion body “myopathy”* rather than “myositis.” Patients with sporadic inclusion body myositis typically respond poorly to steroids and other immunosuppressive drugs used in the treatment of other idiopathic inflammatory myopathies.

Polymyositis

Polymyositis used to be the third inflammatory myopathy in the original triad besides dermatomyositis and inclusion body myositis but cases that were historically classified as polymyositis would now be regarded as immune-mediated necrotizing myopathy, antisynthetase syndrome, or sporadic inclusion body myositis.

Toxic Myopathies and Myopathy With Metabolic Changes

Medications, non-medicinal drug use, and hormonal changes can result in muscle damage causing weakness and CK level elevations. Different mechanisms are described:

1. *Statins, fibrates, and high doses of alcohol* are examples of agents that can cause acute damage and necrosis of individual myofibers. In severe cases, muscle breakdown can result in rhabdomyolysis and associated complications such as acute renal failure. The risk for direct statin toxicity increases with the statin dose, and age over 65 years, renal failure, hepatic failure, and hypothyroidism. Affected patients typically improve with discontinuation of the offending drug and supportive care.

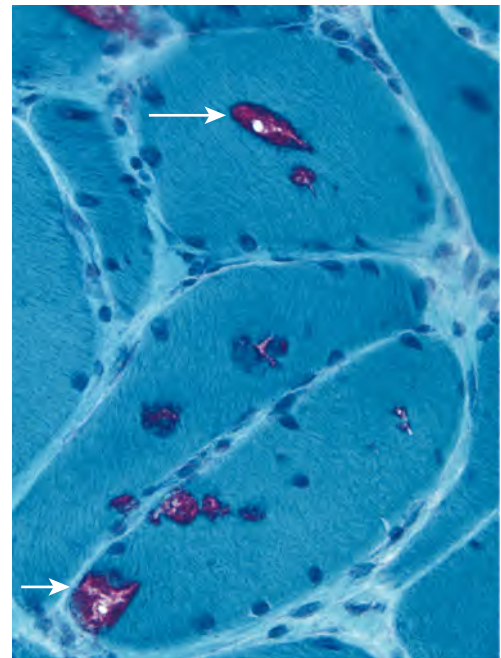


Fig. 27.10 Inclusion body myositis, showing myofibers containing rimmed vacuoles—inclusions with reddish granular rimming (arrows). Modified Gömöri trichrome stain.

2. *Some medications like statins and immune checkpoint inhibitors* can trigger an immune-mediated myopathy. The distinction between pure statin toxicity and IMNM in the context of statin exposure (discussed earlier) is sometimes difficult to make at the time of presentation.
3. *Antiretrovirals like zidovudine* can lead to mitochondrial damage and a form of acquired mitochondrial myopathy (see below).
4. *Medications that interfere with lysosomal function such as hydroxychloroquine and chloroquine* can cause a myopathy of insidious onset that mimics a lysosomal storage disease. The muscle tissue in such cases shows myopathic changes including vacuolation that predominantly affects type I fibers. Ultrastructural studies identify aggregates of whorled, lamellar membranous structures. Cardiac muscle may also be affected by these drugs and can exhibit similar pathologic changes.
5. *Catabolic endogenous or exogenous corticosteroids* can cause selective atrophy of type II fibers and proximal muscle weakness. Critical illness myopathy, also known as myosin-deficient myopathy, is a neuromuscular disorder seen in critically ill patients typically treated in the ICU whose management includes corticosteroid therapy together with depolarizing neuromuscular blocking agents to allow mechanical ventilation. In this setting, there may be relatively selective degradation of sarcomeric myosin thick filaments, producing profound weakness that can complicate the clinical course, for example by interfering with weaning from mechanical ventilation.
6. *Thyroid dysfunction* can lead to several types of myopathy. *Thyrotoxic myopathy* presents most commonly as an acute or chronic proximal muscle weakness that may precede other signs of hyperthyroidism. Such patients may also present with exophthalmic ophthalmoplegia, characterized by swelling of the eyelids, edema of the conjunctiva, and diplopia. Hypothyroidism can cause cramping or aching of muscles. Movement and reflexes may be slowed. Findings in skeletal

muscle include fiber atrophy, an increased number of abnormally localized nuclei, glycogen aggregates, and (occasionally) deposition of mucopolysaccharides in connective tissue.

Inherited Diseases of Skeletal Muscle

Inherited mutations causing defects in proteins needed for skeletal muscle function are responsible for a diverse collection of disorders. In some of these disorders, skeletal muscle is the main site of disease, but in others multiple organs are involved. Of these, the heart is of particular importance since cardiac involvement is common and often life limiting. Channelopathies and metabolic myopathies are also often the result of underlying mutations but will be discussed in separate sections. Genetic neuromuscular diseases have been a major target of efforts to introduce various forms of gene therapy, discussed later.

Historically, inherited myopathies have been subdivided into several broad categories based on inheritance pattern, anatomic pattern of muscle involvement, age of onset, clinical course, and underlying pathogenesis. Included in the group of inherited muscular disorders are:

- *Congenital myopathies* (Table 27.3) typically present in infancy. Some severe forms are associated with early lethality while others may be associated with muscle defects that tend to be static and are associated with a near-normal life span. Congenital myopathies are often associated with distinct structural abnormalities of the muscle.
- *Muscular dystrophies* are characterized by progressive muscle damage that typically comes to attention after infancy (discussed later).
- *Congenital muscular dystrophies* tend to present in infancy and are often associated with developmental abnormalities of the CNS as well as progressive muscle damage. Congenital muscular dystrophies include two important groups:

TABLE 27.3 Congenital Myopathies

Disease and Inheritance	Gene and Locus	Clinical Features	Pathologic Features
Central core disease; autosomal dominant	Ryanodine receptor-1 (<i>RYR1</i>) gene; 19q13.2	Early-onset hypotonia and weakness; “floppy infant”; associated skeletal abnormalities like scoliosis, hip dislocation, or foot deformities; some <i>RYR1</i> mutations cause central core disease, some cause malignant hyperthermia, and some cause both	Cytoplasmic cores represent demarcated central zones in which the normal arrangement of sarcomeres is disrupted and mitochondria are decreased in number
Nemaline myopathy (NEM)	AD NEM1— α -tropomyosin 3 (<i>TPM3</i>) gene; 1q21.3 AR NEM2—nebulin (<i>NEB</i>) gene; 2q23.3 AR NEM3— α -actin-1 (<i>ACTA1</i>) gene; 1q42.13 AD NEM4—tropomyosin-2 (<i>TPM2</i>) gene; 9p13.3 AR NEM5—troponin T1 (<i>TNNT1</i>) gene; 19q13.42 AR NEM7—cofilin-2 (<i>CFL2</i>) gene; 14q13.1	Childhood weakness, sometimes severe with hypotonia at birth (“floppy infant”)	Aggregates of spindle-shaped particles (<i>nemaline rods</i>); occur predominantly in type I fibers; derived from Z-band material (α -actinin); best seen on modified Gömöri stain or by electron microscopy
Centronuclear (myotubular) myopathy	XL—myotubularin (<i>MTM1</i>) gene; Xq28 AD—dynammin-2 (<i>DNM2</i>) gene (and others); 19p13.2 AR—amphiphysin-2 (<i>BIN1</i>) gene; 2q14.3	Severe congenital hypotonia, “floppy infant,” and poor prognosis in X-linked form (“myotubular myopathy”) Childhood onset or young adult onset with other variants with weakness and hypotonia	Many fibers contain nuclei in the geometric center of the myofiber; central nuclei are more common in type I fibers, which are small in diameter, but can occur in both fiber types

AD, Autosomal dominant; AR, autosomal recessive; XL, X-linked.

- **Conditions with defects in extracellular matrix surrounding myofibers.** These are exemplified by Ullrich congenital muscular dystrophy (UCMD) and merosin deficiency. In the former, the causative mutations involve one of three collagen VI α genes; in the case of merosin deficiency, the gene encoding merosin (LAMA2) is disrupted. UCMD is characterized by hypotonia, proximal contractures, and distal hyperextensibility. A morphologic hallmark is spatial separation of the matrix proteins perlecan and collagen VI, which normally co-localize.
- **Conditions with abnormalities in receptors for extracellular matrix.** This group includes diseases that disrupt the post-translational modification of α -dystroglycan (Fig. 27.11) by O-linked glycosylation. Mutations of α -dystroglycan itself typically result in fetal demise, but defects in its posttranslational glycosylation result in milder forms of dystroglycan dysfunction. α -dystroglycan is also important for CNS and eye development, which explains why severe loss of function mutations result in developmental defects of the CNS and eyes as well as congenital muscular dystrophy. Milder forms may cause only skeletal muscle disease. Some of these mutations are also linked to limb-girdle muscular dystrophy (see later).

Muscular Dystrophies

Muscular dystrophies include many inherited disorders marked by progressive skeletal muscle damage that typically manifest between childhood and adulthood. As mentioned earlier, except for congenital muscular dystrophies, these diseases do not present in infancy.

X-Linked Muscular Dystrophy With Dystrophin Mutation Including Duchenne and Becker Muscular Dystrophy. The most common muscular dystrophies are X-linked and stem from mutations that disrupt the function of a large structural protein called *dystrophin*. As a result, these diseases are sometimes referred to as *dystrophinopathies*. The most common early-onset form is *Duchenne muscular dystrophy*. It has an incidence of 1 per 3500 live male births and has a severe progressive phenotype. *Becker muscular dystrophy* is the second relatively common dystrophinopathy; it is characterized by later disease onset and a milder phenotype. Other rare dystrophinopathies may present with isolated cardiomyopathy, asymptomatic elevations of CK levels, or exercise intolerance because of myalgias and cramps. As with many X-linked diseases, female carriers of dystrophin mutations may be mildly symptomatic due to unfavorable X-chromosome inactivation.

Pathogenesis

Duchenne and Becker muscular dystrophies are caused by loss-of-function mutations in the dystrophin gene on the X chromosome. *Dystrophin* is one of the largest human genes, spanning 2.3 million base pairs and containing 79 exons. The encoded protein, dystrophin, is a key component of the dystrophin glycoprotein complex (see Fig. 27.11). This complex spans the plasma membrane and serves as a link between the cytoskeleton inside the myofiber and the basement membrane outside of the cell. By doing so, dystrophin is thought to provide mechanical stability to the myofiber and its cell membrane during muscle contraction. Defects in the complex may lead to small membrane tears that permit influx of calcium, triggering events that result in myofiber necrosis. In addition to its mechanical function, dystrophin may have a role in signaling pathways; for example, its carboxy terminus interacts with nitric oxide synthase, which generates nitric oxide (NO).

Identification and characterization of pathogenic dystrophin mutations has provided an explanation for some of the phenotypic variation in patients with dystrophinopathies. Duchenne muscular dystrophy is typically associated with deletions or frameshift mutations that result

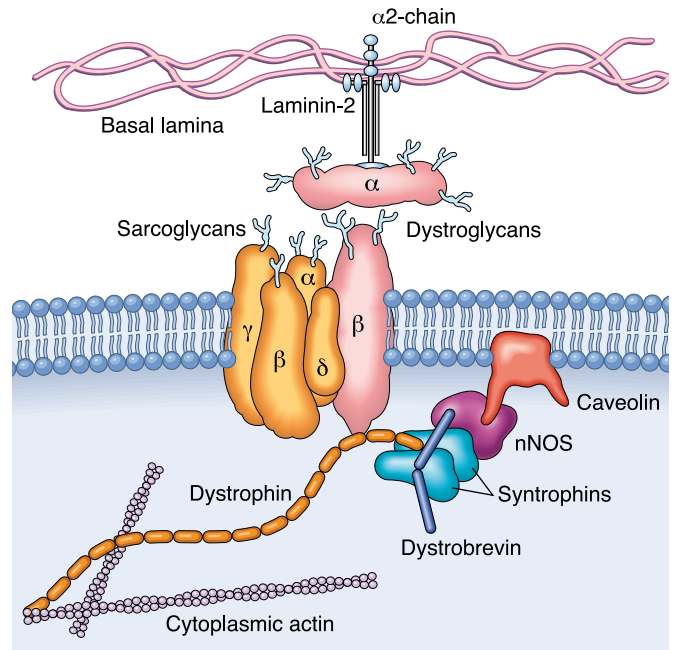


Fig. 27.11 Relationship between the cell membrane (sarcolemma) and the sarcolemmal associated proteins. Dystrophin, an intracellular protein, forms an interface between the cytoskeletal proteins and a group of transmembrane proteins, the dystroglycans and the sarcoglycans. These transmembrane proteins have interactions with the extracellular matrix, including the laminin proteins. Dystrophin also interacts with dystrobrevin and syntrophins, which form a complex with neuronal-type nitric oxide synthetase (*nNOS*) and caveolin. Mutations in dystrophin are associated with the X-linked muscular dystrophies; mutations in caveolin and the sarcoglycan proteins are associated with the limb-girdle muscular dystrophies, which can be autosomal dominant or recessive disorders; and mutations in the α_2 -laminin (merosin) are associated with autosomal recessive congenital muscular dystrophy. Sugar moieties, depicted as light blue branching structures on the dystroglycans, are disrupted by mutations in genes that encode proteins involved in glycosylation.

in total absence of dystrophin. By contrast, the deletions or splicing mutations in Becker muscular dystrophy permit the synthesis of a truncated version of dystrophin that retains partial function.

MORPHOLOGY

The basic morphologic changes that muscular dystrophies elicit in skeletal muscle tissue may differ in severity but are otherwise similar. The changes in Duchenne muscular dystrophy serve as an example. This disease is marked by chronic muscle injury that outpaces the capacity for repair (Fig. 27.12). Muscle biopsies in young boys show ongoing damage in the form of **segmental myofiber necrosis and regeneration associated with an admixture of atrophic myofibers**. The fascicular architecture is preserved during the earlier stages of the disease, and there is usually no inflammation except for the presence of myophagocytosis. As the disease progresses, muscle tissue is replaced by collagen and fat cells ("**fatty replacement**" or "**fatty infiltration**"). This remodeling distorts the fascicular architecture of the muscle, which becomes markedly abnormal over time. At later time points, the remaining myofibers show prominent variation in size ranging from small atrophic fibers to large, hypertrophied fibers. Immunohistochemical studies for dystrophin can be used to document complete or partial absence of dystrophin in Duchenne (Fig. 27.12B) and Becker muscular dystrophy, respectively.

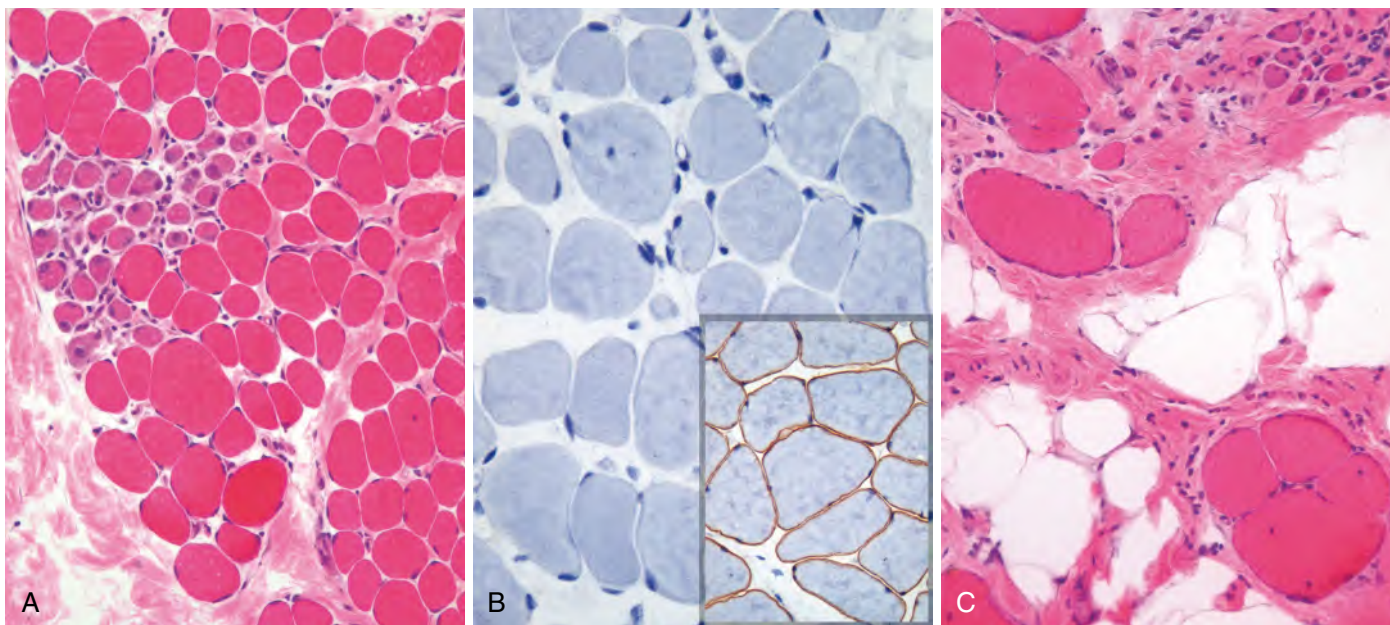


Fig. 27.12 Duchenne muscular dystrophy. Histologic images of muscle biopsy specimens from two brothers. (A–B) Specimens from a 3-year-old boy. (A) At a younger age, fascicular muscle architecture is maintained, but myofibers show variation in size. There is a cluster of basophilic regenerating myofibers (*left side*) and slight endomysial fibrosis, seen as focal pink-staining connective tissue between myofibers. (B) Immunohistochemical staining shows complete absence of membrane-associated dystrophin, stained brown in normal muscle (*inset*). (C) Biopsy specimen from the older (9-year-old) brother illustrates disease progression marked by extensive variation in myofiber size, fatty replacement, and endomysial fibrosis.

Clinical Features

Boys with Duchenne muscular dystrophy appear normal at birth. Very early motor milestones are met, but walking is often delayed. The first indications of muscle weakness are clumsiness and inability to keep up with peers. Weakness begins in the pelvic girdle muscles and then extends to the shoulder girdle. Enlargement of the muscles of the lower leg associated with weakness, termed *pseudohypertrophy*, is often present. The mean age of wheelchair dependence is around 9.5 years. Patients develop joint contractures, scoliosis, worsening respiratory reserve, and sleep hypoventilation as the disease continues to progress.

Dystrophin is also expressed in the heart and the CNS. Dystrophin deficiency in cardiac muscle often leads to the development of cardiomyopathy and arrhythmias, particularly in older patients. Cognitive impairment and learning disabilities, presumably due to a functional role for dystrophin in the brain, are also common and sometimes produce intellectual disability. Patients with Duchenne muscular dystrophy often succumb to their disease before the age of 25 or 30 years because of respiratory insufficiency, pulmonary infections, and heart failure despite optimal supportive care. This contrasts with Becker muscular dystrophy, which typically presents in later childhood, adolescence, or even adult life; has slower progression; and may allow a near-normal life expectancy. Cardiac involvement leading to heart failure is a common life-limiting complication.

The diagnosis is based on history, physical examination, and laboratory studies. Serum creatine kinase is markedly elevated during the first decade of life due to ongoing muscle damage, and then falls as the disease progresses and muscle mass is lost. Genetic testing provides the ultimate confirmation of the diagnosis, but muscle biopsies can still be helpful in cases with difficult to identify genetic alterations.

Limb-Girdle Muscular Dystrophy. Limb-girdle muscular dystrophies are a heterogeneous group of at least 8 autosomal dominant

and 23 autosomal recessive entities. Their overall incidence is 1 in 25,000 to 50,000 individuals. As indicated by the name, all forms are characterized by muscle weakness that preferentially involves proximal muscle groups. Both age of onset and disease severity are highly variable. The causative mutations involve genes that participate in diverse cellular functions, making it difficult to discern a unifying mechanism of disease pathogenesis. Based on current knowledge, the implicated genes can be grouped according to the proteins they encode:

- *Structural components* of the dystrophin glycoprotein complex such as the sarcoglycans (Fig. 27.11)
- *Enzymes that glycosylate* α -dystroglycan, a component of the dystrophin glycoprotein complex (see Fig. 27.11)
- *Proteins that associate with the Z-disks of sarcomeres*
- *Proteins involved in vesicle trafficking and cell signaling*
- Proteins of seemingly diverse function such as the protease calpain 3, and lamin A/C (which is also mutated in some patients with Emery-Dreifuss muscular dystrophy; see next)

Emery-Dreifuss Muscular Dystrophy (EMD). EMD is caused by mutations in genes that encode nuclear lamin proteins. Clinically, it is marked by a triad consisting of slowly progressive humeroperoneal weakness; cardiomyopathy associated with conduction defects; and early contractures of the Achilles tendon, spine, and elbows. The X-linked form (EMD1) and the autosomal form (EMD2) are caused by mutations in the genes encoding emerin and lamin A/C, respectively, both of which localize to the inner face of the nuclear membrane. It is hypothesized that these proteins help maintain the shape and mechanical stability of the nucleus during muscle contraction. They may also influence gene expression by affecting chromatin organization in the nucleus. How defects in these proteins produce the observed phenotypes is unknown.

Facioscapulohumeral Dystrophy. Facioscapulohumeral dystrophy is associated with a characteristic pattern of muscle involvement that includes prominent weakness of facial muscles and muscles of the shoulder girdle. It is an autosomal dominant disease affecting about 1 in 20,000 individuals.

Pathogenesis

The pathogenesis of facioscapulohumeral dystrophy is complex and unique. The disease is linked to epigenetic derepression of the gene encoding the transcription factor *DUX4*, which is silenced in most somatic tissues and can be toxic in cells such as myofibers. In most cases the derepression is thought to result from mutations in regulatory sequences that lead to altered DNA methylation, chromatin relaxation, and *DUX4* transcription in skeletal muscle. In a small proportion of cases the derepression of *DUX4* results from mutations in chromatin repressors such as *SMCHD1*.

Ion Channel Myopathies (Channelopathies)

Channelopathies are a group of inherited diseases caused by mutations affecting the function of ion channel proteins. Most channelopathies are autosomal dominant disorders with variable penetrance. Depending on the channel that is affected, clinical manifestations may include epilepsy, migraine, movement disorders with cerebellar dysfunction, peripheral nerve disease, and muscle disease. The following discussion focuses on those channelopathies with muscle involvement.

Myotonic Dystrophy. Myotonic dystrophy is an autosomal dominant multisystem disorder associated with skeletal muscle weakness, cataracts, endocrinopathy, and cardiomyopathy. It affects about 1 in 10,000 individuals. Myotonia, a sustained involuntary contraction of muscles, is a key feature of this disease, which can be variably classified with muscular dystrophies or channelopathies based on its unique pathogenesis.

Pathogenesis

Myotonic dystrophy is a trinucleotide repeat expansion disease that results from CTG expansion in the *DMPK* gene (type 1, DM1) or CCTG expansions in the *CNBP* gene (type 2, DM2). The following discussion focuses on the more common DM1 variant. The correlation between the length of expansion and disease severity is not as strong as in some other triplet repeat expansion disorders like Huntington disease (Chapter 28) but severe congenital myotonic dystrophy is typically linked to larger repeat sizes. Experimental studies suggest that the skeletal muscle phenotype stems from a “toxic” gain-of-function caused by the triplet repeat expansion. Specifically, the expanded CTG-repeats in the *DMPK* mRNA transcript appear to bind and sequester muscle splicing regulators. This, in turn, is thought to lead to alternative splicing of several genes including *CLCN1*, which encodes a chloride channel. It is believed that the resulting deficiency of this chloride channel is responsible for the myotonia; of note in this regard, mutations in *CLCN1* itself are linked to congenital myotonia. Insulin resistance and diabetes in DM1 may be the result of alternate splicing of insulin receptor isoforms in skeletal muscle. Differential expression of other genes and accumulation of transcripts containing the trinucleotide repeat may contribute to other manifestations of this disease.

Other Ion Channel Myopathies. Different ion channel myopathies may cause decreased or increased excitability resulting in hypotonia or hypertonia. Disorders associated with hypotonia can be further subclassified based on whether symptomatic patients have elevated,

depressed, or normal serum potassium levels and are called *hyperkalemic*, *hypokalemic*, or *normokalemic periodic paralysis*, respectively. Examples of genes that when mutated lead to periodic paralysis or related disorders include the following:

- *KCNJ2*: Mutations affecting this potassium channel cause *Andersen-Tawil syndrome*, an autosomal disorder associated with periodic paralysis, cardiac arrhythmias, and skeletal abnormalities.
- *SCN4A*: Mutations affecting this sodium channel cause several autosomal disorders with presentations ranging from myotonia to periodic paralysis.
- *CACNA1S*: Missense mutations in this protein, a subunit of a muscle calcium channel, are the most common cause of hypokalemic paralysis.
- *CLCN1*: Mutations affecting this chloride channel cause myotonia congenita. As already discussed, *CLCN1* expression is decreased in myotonic dystrophy.
- *RYR1*: Mutations in the *RYR1* gene disrupt the function of the ryanodine receptor, which regulates calcium release from the sarcoplasmic reticulum. *RYR1* mutations are linked to a congenital myopathy (central core disease) and to *malignant hyperthermia*. The latter is characterized by a hypermetabolic state (tachycardia, tachypnea, muscle spasms, and later hyperpyrexia) that can be triggered by anesthetics, most commonly halogenated inhalational agents and succinylcholine. Upon exposure to anesthetic, the mutated receptor allows increased efflux of calcium from the sarcoplasmic reticulum, leading to tetany and excessive heat production.

Diseases of Lipid or Glycogen Metabolism

Many inborn errors of lipid or glycogen metabolism affect skeletal muscle. These disorders tend to produce one of two general patterns of muscle dysfunction. Some lead to patients presenting with episodic symptoms that are brought on by events like fasting or exercise and can include muscle cramping, pain, or rhabdomyolysis from extensive muscle necrosis. Other metabolic myopathies result in slowly progressive damage akin to a dystrophy without episodic manifestations. Listed here are some examples:

- *Carnitine palmitoyltransferase II deficiency* is the most common disorder of lipid metabolism to cause episodic muscle damage with exercise or fasting. The defect in this disorder impairs the transport of free fatty acids into mitochondria.
- *Myophosphorylase deficiency (McArdle disease)* is one of the more common glycogen storage diseases affecting skeletal muscle; it also results in episodic muscle damage with exercise.
- *Acid maltase deficiency* results in impaired lysosomal metabolism of glycogen, glycogen aggregation, and disruption of lysosomal function. Severe deficiency results in the generalized glycogenosis of infancy, *Pompe disease* (Chapter 5). Milder deficiency can cause a progressive adult-onset myopathy that preferentially involves the respiratory and truncal muscles. Enzyme replacement therapy has been used to treat some affected patients and clinical trials are ongoing for possible gene replacement therapy.

Mitochondrial Myopathies

Mitochondrial diseases are systemic conditions that can affect many organ systems, including skeletal muscle. Many of the causative mutations impair the ability of mitochondria to generate adenosine triphosphate (ATP). The genetics of these disorders are varied and complex.

Pathogenesis

Mitochondrial proteins and tRNAs may be encoded by either the nuclear genome or the mitochondrial genome (mtDNA). While

mutations in nuclear mitochondrial genes follow Mendelian inheritance patterns, mutations in mtDNA are maternally inherited, since all of the mitochondria in the embryo are contributed by the oocyte (Chapter 5). In addition, unlike nuclear DNA, which is present in only two copies and is evenly distributed from a mother cell to daughter cells during cell division, each cell contains thousands of mtDNA copies, which are distributed in a random fashion to daughter cells at the time of cell division resulting in heteroplasmy. It is believed that disease results only when a certain threshold of mutated mtDNA copies is exceeded within a substantial fraction of “at-risk” cells (e.g., skeletal muscle cells) in a tissue.

MORPHOLOGY

The most consistent pathologic change in skeletal muscle is **abnormal aggregates of mitochondria** that are seen preferentially in the subsarcolemmal area of affected myofibers, producing an appearance that is referred to as **ragged red fibers** (Fig. 27.13). By electron microscopy, morphologically abnormal mitochondria may be seen. Loss of particular mitochondrial enzyme activities characterizes some mitochondrial diseases and may be appreciated by histochemical staining for cytochrome oxidase. Some mitochondrial diseases lack morphologic changes and can be diagnosed only through enzymatic assays or genetic analyses.

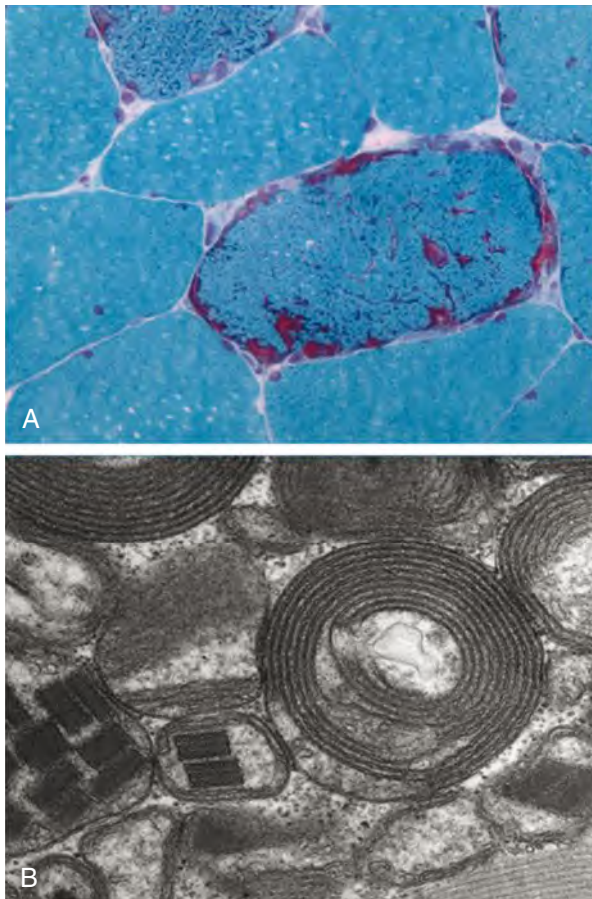


Fig. 27.13 Mitochondrial myopathy. (A) Ragged red fiber with reddish granular subsarcolemmal staining reflective of abnormal aggregation of mitochondria (Gomori trichrome stain). (B) Electron micrograph showing morphologically abnormal mitochondria with concentric membranous rings (so-called “phonograph records”) and rhomboid paracrystalline inclusions (lower left).

Clinical Features

Skeletal muscle involvement can manifest in many ways including weakness, elevations in serum creatine kinase levels, or rhabdomyolysis. Although the anatomic pattern of muscle weakness is variable, involvement of extraocular eye muscles is common and can be a clue to the diagnosis. Indeed, *chronic progressive external ophthalmoplegia* is a common feature of mitochondrial disorders and may occur as an isolated phenomenon or a part of a multisystem syndrome. The reason that extraocular eye muscles are particularly sensitive to mitochondrial disease is uncertain, but it may be that these muscles have exceptionally high requirements for ATP. In line with this idea, extraocular eye muscles have the most mitochondria per mass of any of the body’s muscles.

Due to the complexity of mitochondrial genetics, genotype/phenotype relationships in mitochondrial disorders are not straightforward. For example, a single point mutation in the mitochondrial leucine tRNA gene may produce isolated chronic progressive external ophthalmoplegia in one patient and, in another, a much more severe phenotype, *mitochondrial encephalomyopathy with lactic acidosis and stroke-like episodes* (MELAS). Similarly, deletions in mtDNA may lead to either isolated ophthalmoplegia or to *Kearns-Sayre syndrome*, characterized by ophthalmoplegia, pigmentary degeneration of the retina, and complete heart block. *Myoclonic epilepsy with ragged red fibers* and *Leber hereditary optic neuropathy* are other examples of mitochondrial disease caused by point mutations in mtDNA. Many mitochondrial disorders, such as subacute necrotizing encephalopathy (*Leigh syndrome*), are remarkably heterogeneous genetically and may be caused by mutations in either mtDNA or the nuclear genome. In the case of Leigh syndrome, causative mutations have been identified in more than 30 different genes, the common feature being that all of the affected genes encode proteins with essential roles in mitochondrial metabolism.

Gene Therapy for Neuromuscular Diseases

Neuromuscular diseases have been a focus of efforts to develop genetic treatment strategies. These must be tailored to the pathogenesis of the specific disease process and in some instances the specific mutation. Loss-of-function mutations may be treated with gene replacement therapy or gene modulation.

- Adeno-associated virus (AAV) has been under investigation to deliver DNA constructs that can replace defective genes. Limitations include a limited packaging capacity of about 4.7 kilobases (the *DMD* gene, for example, has 14 kb of coding sequence). Nonetheless, an AAV vector engineered to deliver a “mini-dystrophin” gene has recently been approved for treatment of boys with Duchenne muscular dystrophy. The long-term benefits of such an approach remain to be determined. Other limiting factors are naturally occurring antibodies neutralizing AAV as well as an immune response elicited by viral proteins or by the transgene that is newly expressed in the target tissue. Hepatotoxicity is also a common side effect with AAV gene therapy.
- Strategies for gene modulation under investigation or in use for some patients with Duchenne muscular dystrophy are exon skipping using antisense oligonucleotides or drugs promoting ribosomal read-through of nonsense mutations. Exon skipping using antisense oligonucleotides is a mutation-specific approach and is currently available for patients with variants in exons 45, 51, or 53 of the *dystrophin* gene that disrupt the normal reading frame. In these patients, exon skipping results in the production of a shortened but partly functional protein that can ameliorate the clinical phenotype. Medications promoting ribosomal read-through have achieved approval in Europe for those 10% to 15% of Duchenne patients with nonsense mutations.

- Knockdown strategies offer a possible approach for patients with dominantly inherited diseases linked to possible toxic gain-of-function and may utilize RNA interference or gene silencing with antisense oligonucleotides. Still farther from clinical implementation are gene correction therapies utilizing CRISPR/Cas9 or similar technologies.

KEY CONCEPTS

Disorders of Skeletal Muscle

- Altered muscle function may stem from neurogenic or primary myopathic processes.
- Myopathic disorders are often marked by necrosis and regeneration of myofibers.
- Inflammatory myopathies are often linked to distinct autoantibodies, clinical presentation, and histopathologic changes and include dermatomyositis (DM), antisynthetase syndrome, immune-mediated necrotizing myopathy (IMNM), sporadic inclusion body myositis (sIBM), as well as overlap myositis. This classification has largely replaced the traditional triad of polymyositis, dermatomyositis, and inclusion body myositis.
- DM occurs in children and adults. Distinct clinical subtypes are associated with specific autoantibodies. DM can be a paraneoplastic disease in adults. A hallmark feature not found in all cases is perifascicular atrophy.
- Antisynthetase syndrome is associated with distinct antibodies, most often anti-Jo1, and necrotizing changes in a perifascicular distribution.
- IMNM is associated with HMGR or SRP antibodies, is associated with muscle necrosis, and often lacks prominent inflammatory cell infiltrates.
- sIBM is a chronic, progressive disease of older patients associated with rimmed vacuoles.
- Muscular dystrophies and congenital myopathies result from genetic mutations that disrupt the function of proteins that are important for various aspects of muscle development, function, and regeneration. Some of these diseases present in infancy, others in adulthood. They may be relentlessly progressive or cause relatively static deficits.
- Myopathy can result from toxic injury or be the result of metabolic diseases including those of lipid metabolism, glycogen metabolism, and mitochondria.

PERIPHERAL NERVE SHEATH TUMORS

Most benign and malignant neoplasms of peripheral nerve sheaths are composed of cells that show evidence of Schwann cell differentiation. These include three common types: schwannoma, neurofibroma, and malignant peripheral nerve sheath tumor (MPNST). Peripheral nerve sheath tumors have several unique features. First, they are associated with relatively common familial tumor syndromes, including neurofibromatosis type 1 (NF1), neurofibromatosis type 2 (NF2), and schwannomatosis. Second, MPNSTs seen in the context of NF1 are thought to arise through malignant transformation of preexisting neurofibromas. Malignant transformation of a preexisting benign lesion is a common origin for certain carcinomas (e.g., colon cancer) but is unusual in soft tissue tumors.

Neurofibromatosis

Neurofibromatosis Type 1

This is a common autosomal dominant disorder with a frequency of 1 in 3000 individuals. It is a systemic disease associated with nonneoplastic manifestations and with a variety of tumors, including neurofibromas, MPNSTs, gliomas of the optic nerve, other gliomas (Chapter 28), and pheochromocytomas. Other features include

intellectual disability, seizures, skeletal defects, pigmented nodules of the iris (*Lisch nodules*), and cutaneous hyperpigmented macules (*café au lait spots*). The disease is caused by loss-of-function mutations in the *NF1* gene, located at 17q11.2, which encodes the tumor suppressor neurofibromin. The neoplastic cells in NF1-related tumors lack neurofibromin due to biallelic defects of the *NF1* gene. As discussed in Chapter 7, NF1 protein has GTPase activity that restrains RAS function. In the absence of NF1, RAS remains trapped in its active state.

The disease has a high penetrance but variable expressivity. Some patients exhibit only subtle features, while others show disease that is restricted to certain parts of the body, a distribution that is attributable to mosaicism. An unfortunate subset has severe disease. Large chromosomal deletions that span *NF1* and extend to involve adjacent genes tend to be associated with more severe phenotypes.

Neurofibromatosis Type 2

This is an autosomal dominant disorder associated with the development of schwannomas, meningiomas, and spinal ependymomas (but not neurofibromas). Bilateral vestibular schwannomas are a hallmark of this disease and are found in many affected patients. This disorder is less common than NF1, having a frequency of 1 in 40,000 to 50,000 individuals. Certain other rare familial syndromes are also associated with multiple schwannomas, such as schwannomatosis.

The *NF2* gene is located on chromosome 22q12 and is also commonly mutated in sporadic spinal ependymomas, meningiomas and schwannomas. The *NF2* gene product, merlin, is a cytoskeletal protein that participates in the regulation of several key signaling pathways that are involved in control of cell shape, cell growth, and cell adhesion. There is some correlation between the type of mutation and clinical symptoms, with nonsense and frameshift mutations causing more severe phenotypes than missense mutations.

Schwannomas

These are benign tumors that exhibit Schwann cell differentiation and arise from peripheral nerves. Schwannomas are a component of NF2, and even sporadic schwannomas are commonly associated with somatic inactivating mutations in the *NF2* gene.

MORPHOLOGY

Schwannomas are well-circumscribed, encapsulated masses that attach loosely to the associated nerve without invading it. Grossly, these tumors form firm, gray masses. Microscopically, they are composed of Schwann cells that typically have a spindled elongated nucleus with a wavy or buckled shape. Like normal Schwann cells, the tumor cells express the markers S100 and SOX10 and are surrounded by continuous basement membrane that can be visualized by electron microscopy or immunohistochemistry. Schwannomas typically include regions referred to as **Antoni A areas** in which the cells are arranged in distinct patterns like streaming fascicles and other areas referred to as **Antoni B areas** that appear more myxoid and exhibit less distinct architectural patterning (Fig. 27.14A). Palisading of nuclei is common in the Antoni A areas and arrangements with palisading nuclei around central “nuclear-free zones” are termed **Verocay bodies** (Fig. 27.14B). A variety of degenerative changes may be found in schwannomas, including nuclear pleomorphism, xanthomatous change, vascular hyalinization, cystic change, and necrosis. Some large, mitotically active schwannomas lacking Antoni B areas can mimic a sarcoma.

Clinical Features

Schwann cells are present around the axons in cranial nerves II through XII, nerve roots, large named nerves as well as tiny, distal nerve branches, and schwannomas can occur in all these locations. The

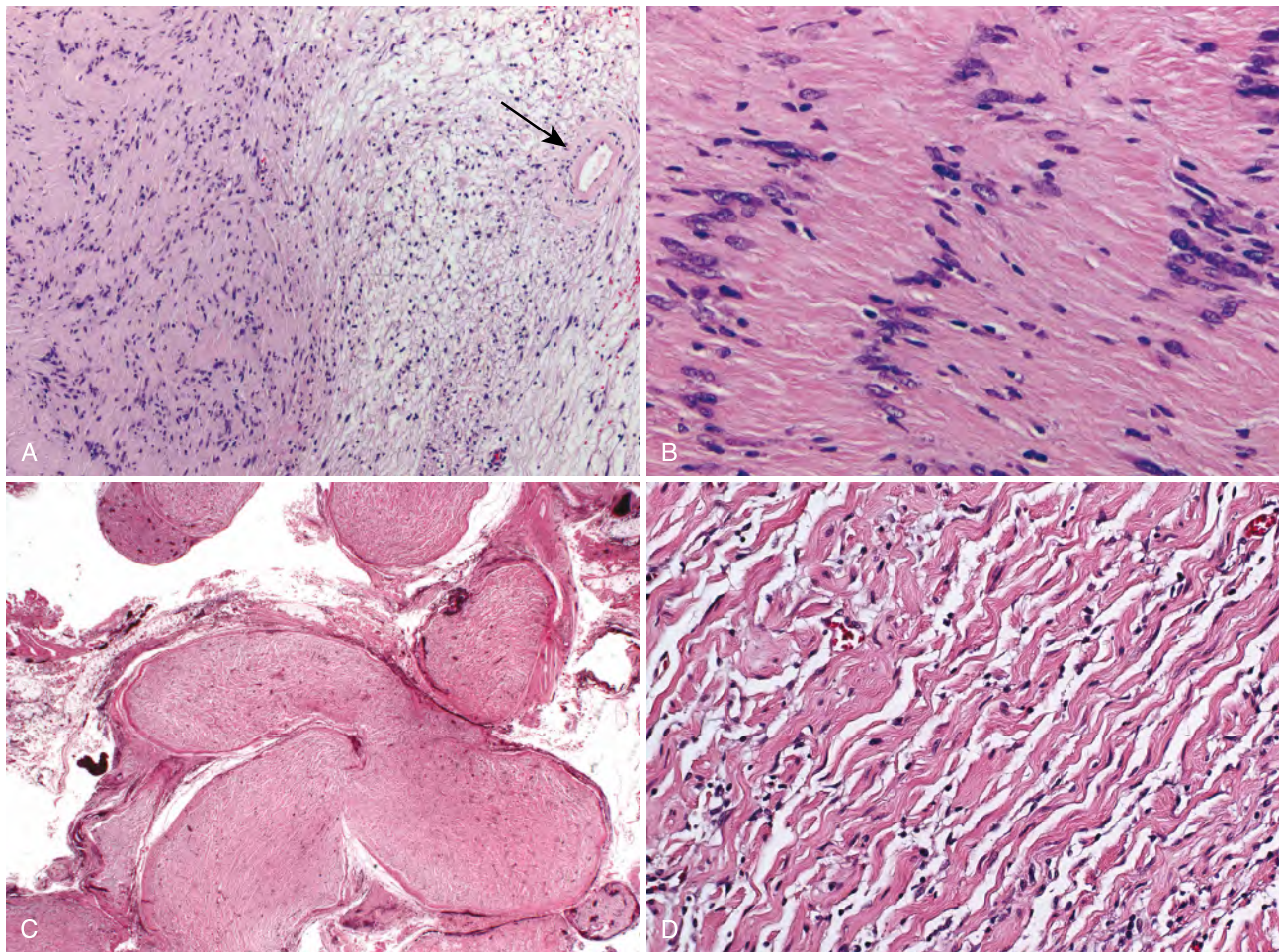


Fig. 27.14 Schwannoma and plexiform neurofibroma. (A and B) Schwannoma. (A) Schwannomas often contain dense eosinophilic Antoni A areas (*left*) and loose, pale Antoni B areas (*right*), as well as hyalinized blood vessels (*arrow*). (B) Antoni A area with the tumor cell nuclei aligned in palisading rows leaving anuclear zones and resulting in the formation of structures termed Verocay bodies. (C and D) Plexiform neurofibroma. (C) Multiple nerve fascicles are expanded by infiltrating tumor cells. (D) At high power, bland spindle cells are admixed with wavy collagen bundles resembling carrot shavings.

presenting symptoms can simply be those of a mass lesion, resulting from compression of the adjacent nerve, or may be related to compression of adjacent CNS tissue. Within the cranial vault, most schwannomas occur at the cerebellopontine angle, where they are attached to the vestibular branch of cranial nerve VIII. Affected individuals often present with tinnitus and hearing loss; the tumor has previously been referred to as an *acoustic neuroma*—a double misnomer, since the tumor neither arises from the acoustic portion of the nerve nor is it a neuroma. A more recently proposed and preferred name is *vestibular schwannoma*. Schwannomas typically grow as a mass abutting but not infiltrating a preexisting nerve. Surgical resection is therefore often possible and can be curative. Stereotactic radiosurgery is another treatment option. Schwannomas may recur locally if incompletely resected, but malignant transformation is extremely rare.

Neurofibromas

Neurofibromas are benign nerve sheath tumors that are more heterogeneous in composition than schwannomas. The neoplastic Schwann cells are admixed with perineurial-like cells, fibroblasts, mast cells, and CD34+ spindle cells. Neurofibromas may be either sporadic or NF1-associated. Different types of neurofibroma can be distinguished depending on their growth pattern.

- *Superficial cutaneous neurofibromas* often present as pedunculated nodules that can be seen isolated (if sporadic) or multiple (if NF1-associated).
- *Diffuse neurofibromas* often present as a large plaque-like elevation of skin and are typically NF1-associated.
- *Plexiform neurofibromas* can be found in deep or superficial locations in association with nerve roots or large nerves and are uniformly NF1-associated.

Pathogenesis

Only the Schwann cells in neurofibromas show complete loss of the NF1 gene product, neurofibromin, indicating that these are neoplastic cells. Haploinsufficiency for the NF1 gene in other associated cells may also contribute to the growth of NF1-associated tumors. For example, there is evidence that NF1-haploinsufficient mast cells are hypersensitive to KIT ligand produced by Schwann cells and in response secrete factors that stimulate Schwann cell growth. This form of tumor/stromal cell cross-talk may be targetable with inhibitors of the KIT receptor tyrosine kinase. Other studies suggest that plexiform neurofibromas and dermal neurofibromas arise from different neural crest-derived precursor cells. Malignant transformation of neurofibroma to MPNST is usually observed in the plexiform variant but is sometimes also seen

in the diffuse type. The overall incidence of MPNST in NF1 patients is about 5% to 10%, but patients with large numbers of plexiform neurofibromas and large deletions in the *NF1* gene are at higher risk.

MORPHOLOGY

Localized cutaneous neurofibroma. These are small, well-delineated but unencapsulated nodular lesions that arise in the dermis and subcutaneous fat. They have relatively low cellularity and contain bland Schwann cells admixed with stromal cells such as mast cells, perineurial cells, CD34+ spindle cells, and fibroblasts. Adnexal structures are sometimes entrapped at the edges of the lesion. The stroma of these tumors contains loose collagen.

Diffuse neurofibroma. This tumor has morphologic features similar to those seen in localized cutaneous neurofibromas but exhibits a distinctly different growth pattern. The tumor diffusely infiltrates dermis, subcutaneous connective tissue, or deep soft tissues, entrapping fat and appendage structures, resulting in large, often plaque-like, lesions. Focal collections of cells mimicking the appearance of Meissner corpuscles (so-called **pseudo-Meissner corpuscles** or **tactile-like bodies**) are an associated feature.

Plexiform neurofibroma. These tumors grow within and expand multiple nerve fascicles (Fig. 27.14C), entrapping associated axons. The external perineurial layer of the nerve is preserved, giving individual nodules an encapsulated appearance. The expanded, ropy thickening of multiple nerve fascicles results in what is sometimes referred to as a “bag of worms” appearance upon gross examination. The tumor has a cellular composition similar to that of other neurofibromas. The extracellular matrix varies from loose and myxoid to more collagenous and fibrous. Collagen is often seen in bundles likened to “shredded carrots” (Fig. 27.14D).

Malignant Peripheral Nerve Sheath Tumors

Most malignant peripheral nerve sheath tumors (MPNSTs) are high-grade tumors, but low-grade variants are recognized. About half arise in NF1 patients and are assumed to result from malignant transformation of a plexiform neurofibroma. Sporadic cases may arise de novo. Most are associated with larger peripheral nerves in the chest, abdomen, pelvis, neck, or limb-girdle. MPNSTs exhibit complex chromosomal aberrations including chromosome gains and losses. High-grade MPNSTs exhibit genetic or epigenetic abnormalities in addition to *NF1* alterations. These include loss of the polycomb repressor complex PRC2, which is responsible for placing repression marks on histone H3; and trimethylated histone 3; homozygous loss of *CDKN2A/B*, which encodes several tumor suppressor proteins, including the cell cycle regulator p16 and the p53 regulator p14.

MORPHOLOGY

The lesions are poorly defined tumor masses that infiltrate along the axis of the parent nerve and invade adjacent soft tissues. A wide range of histologic appearances can be encountered. Typical cases show a fasciculated arrangement of spindle cells. At low power the tumor often appears “marbled” due to variations in cellularity. Mitoses, necrosis, and nuclear anaplasia are common. An interesting phenomenon observed in MPNST is described as “divergent differentiation.” This term refers to the presence of focal areas that exhibit other lines of differentiation, including glandular, cartilaginous, osseous, or rhabdomyoblastic morphology. A tumor exhibiting the latter is referred to as **Triton tumor**. Due to the poorly differentiated nature of MPNST, the distinction from an undifferentiated sarcoma may not be straightforward. Helpful clues include a diagnosis of NF1 in the affected patient, a clearly demonstrated anatomic relationship to a nerve or to a preexisting neurofibroma, and the presence of molecular changes described above.

Other Types of Nerve Sheath Tumors

There are other tumors that are thought to be derived from Schwann cells or perineurial cells.

- **Malignant melanotic nerve sheath tumor (MMNST)** was formerly referred to as “melanotic schwannoma” but is driven by distinctly different genetic changes, particularly *PRKARIA* mutations that lead to increased cAMP-dependent signaling. MMNSTs can be associated with Carney complex and are clinically more aggressive than schwannomas.
- **Granular cell tumor** is composed of polygonal cells filled with lysosomes resulting in the appearance of distinctly granular cytoplasm. Most of these form small superficial or mucosal masses. These tumors are associated with genetic changes in genes affecting vacuolar ATPase function.
- **Perineurioma** is a soft tissue lesion that is thought to arise from the perineurial cells that normally surround individual nerve fascicles.

KEY CONCEPTS

Peripheral Nerve Sheath Tumors

- The three common peripheral nerve sheath tumors—schwannoma, neurofibroma, and MPNST—all arise from cells of Schwann cell lineage.
- Schwannomas are encapsulated benign tumors that can be associated with NF2.
- Neurofibromas are benign peripheral nerve sheath tumors sometimes associated with NF1 that can be subtyped as localized cutaneous, diffuse, or plexiform.
- MPNSTs can be de novo sporadic neoplasms or NF1-associated tumors arising through malignant transformation of a (plexiform) neurofibroma.

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